

2025 Chipathon Postmortem Review Analog/AI Track: Onchip Team

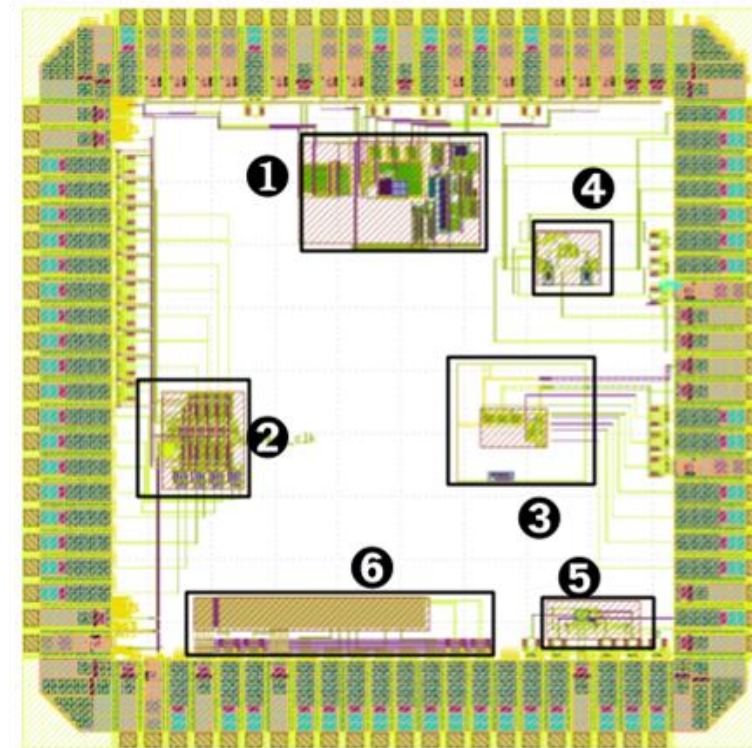
IEEE Solid-State Circuits Society
Technical Committee on the Open Source Ecosystem (TC-OSE)
November 20th, 2025



Analog/AI Track Project Checklists

- A1_AutoMOS
- A2_GenYZ Team
- A3_Mahowalders
- A4_Silicon Souls
- A5_Time Transcenders
- **Axx_Onchip**

Track A



- ❶ AutoMOS:
 - VCO (1.8V, 1GHz)
 - LDO (5V $\rightarrow$ 1.8V, 1mA)
 - Bandgap (1.2V)
- ❷ GenYZ:
 - Instrumentation Amplifier
 - Clock Divider
 - Chopper Switches
 - Low-Pass Filter
- ❸ Mahowalders:
 - Collection of Current Mirrors ($I_{out} = 1.6\mu A$)
- ❹ Silicon Souls:
 - Ring VCO (0.45-2.25 GHz)
- ❺ Time transcenders:
 - Gilbert Mixer ($V_{REF} = 1.2V DC + 0.1V AC$ @89.7MHz, $V_{LO} = 1.8V DC + 0.4V AC$ @100MHz)
- ❻ Onchip
 - Current references (n-type output: $I_{out} = 2\mu A$ and p-type output: $2\mu A$)
 - Folded cascode (DC gain: 95.5[dB], $U_{GB} = 8[MHz]$)

Schedule

- ▶ **Overview** : Final status and progress tracker
- ▶ **Schematic Design** : Design methodology implemented with Xschem (Summary)
- ▶ **CACE Validation** : Verification methodology implemented with CACE (Summary)
- ▶ **Automated Layout** : Layout Design implemented with Glayout flow (Summary)
- ▶ **Conclusions and Comments** : Lessons learned, experiences and suggestions.

Overview

The GitHub Repository

AlexMantilla1/ Onchip_CurRefGen



This repo compiles and shows the work for the OnChip Current Reference Generator in OpenSource tools and PDK



4

Contributors



0

Issues



0

Stars



0

Forks



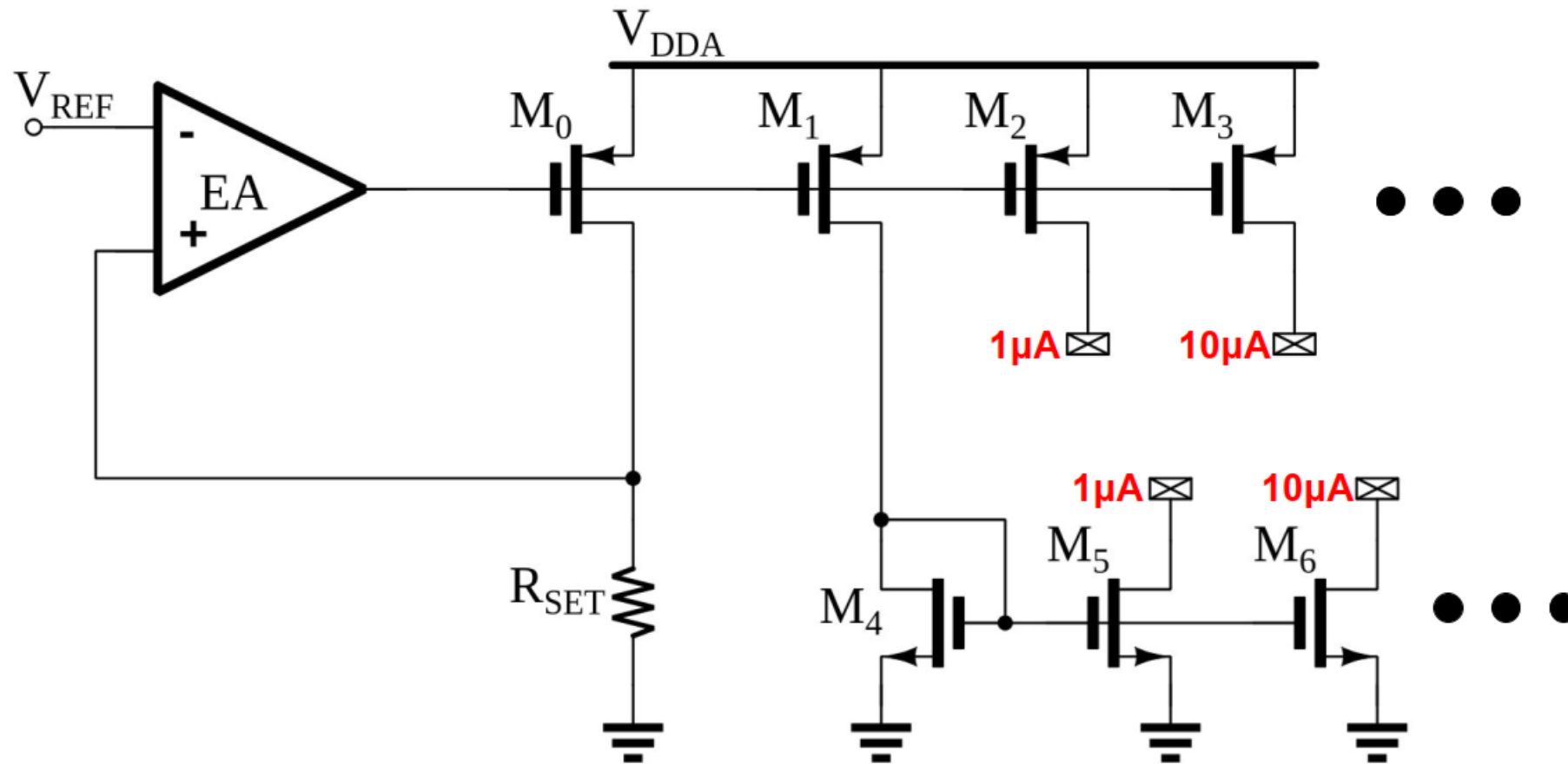
GITHUB.COM

GitHub - AlexMantilla1/Onchip_CurRefGen: This repo compiles and shows the work for the OnChip Current Reference Generator in



https://github.com/AlexMantilla1/Onchip_CurRefGen

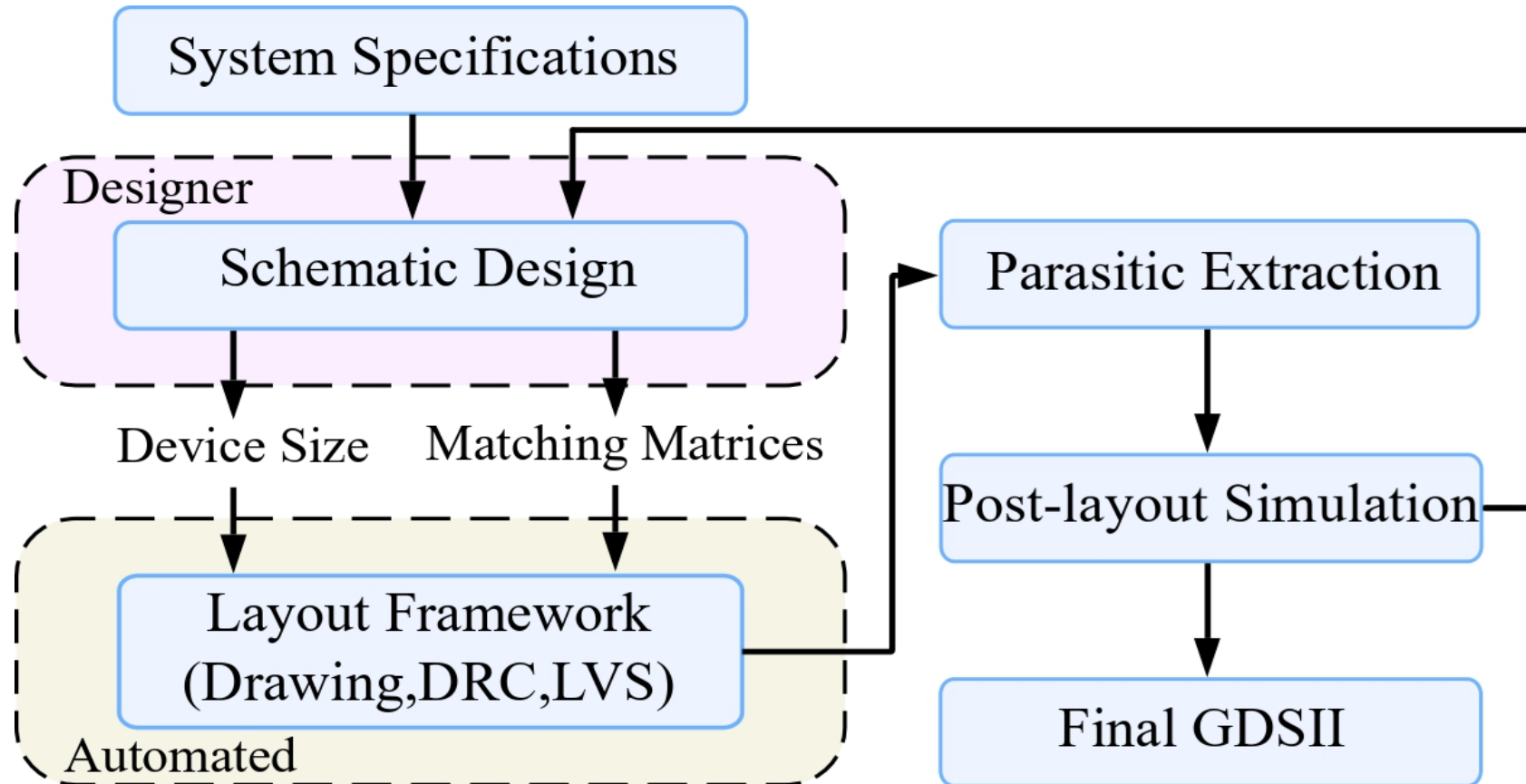
Circuit Built: A Programmable Current Reference Generator



Project Tracker: Sent for Tapeout

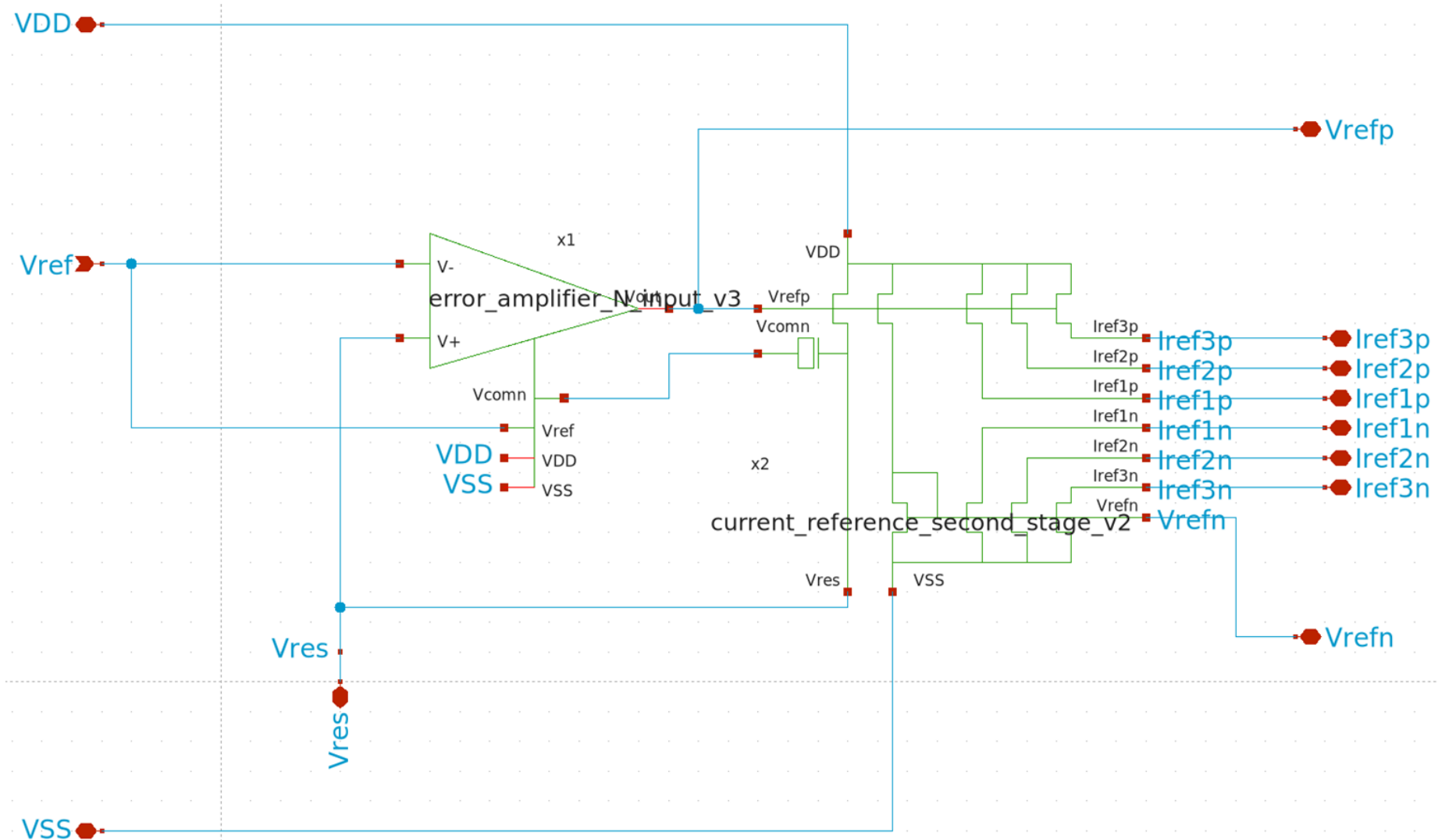
Project Stage	Status	#	Progress (%)	Assigned To	Notes
Planification & Kick-off	Completed		100	Alex Mantilla	Goal: programmable output currents.
ErrAmp Schematic Design	Completed		100	Julian Romero	Prioritized: Offset and Gain. Sch made thinking on layout matching
ErrAmp Schematic VarAware Verification	Completed		100	Julian Romero & Ramón Samiento	Documented CACE flow
ErrAmp Automated Layout	Completed		100	Sebastian Suárez & Jose Algarín	Bias circuit area can be improved
ErrAmp DRC and LVS clean	Completed		100	Sebastian Suárez & Jose Algarín	DRC and LVS clean. Dummies added to xschem sch
ErrAmp post-layout Verification	Completed		100	Julian Romero & Ramón Samiento	Post-layout specs between the limits. Offset improved, but UGBW and gain reduced just a little.
Current Reference Schematic Design	Completed		100	Julian Romero	Indirect compensation used with moscaps, as bigger as better. Mirrors recommended to used same W/L when autogenerated. 600kOmhs External resistor required, which mainly determines the temperature behaviour.
Current Reference VarAware Verification	Completed		100	Julian Romero & Ramón Samiento	Documented CACE flow
Current Reference Automated Layout	Completed		100	Sebastian Suárez & Jose Algarín	Dummies added to xschem sch. Variable output current achieved, specified as input parameter. moscap can increase to fill area if output currents are quite high (not tested).
Current Reference DRC and LVS clean	Completed		100	Sebastian Suárez & Jose Algarín	DRC and LVS clean. Dummies added to xschem sch. No dummies used for compesation caps
Current Reference post-layout Verification	Completed		100	Julian Romero & Ramón Samiento	Simulations are quite heavy. Specs under limits.
Layout Integration to 2nd ESD protection	Completed		100	Alex Mantilla	2nd esd schs shared by vasil.
DRC & LVS clean with 2nd ESD	Completed		100	Alex Mantilla	Area of the metal to diode increased just in case.
Layout Integration to PADring	Completed		100	Alex Mantilla	Merged to AutoMOS GitHub repo.
DRC clean at chip level	Completed		100	Alex Mantilla	After commercial tools validation no DRC errors found.
LVS clean at chip level	In Progress		30	Alex Mantilla	Working on integrating all schematics for chip project.
Documentation	Completed		100	All the Team	GitHub repo documented. https://github.com/AlexMantilla1/Onchip_CurRefGen
Apply to Code a Chip grant	Completed		100	All the Team	Applied on dates.
Contribute to Glayout repo	Not Started		0	All the Team	We really need to clean the code.

Proposed Design Flow

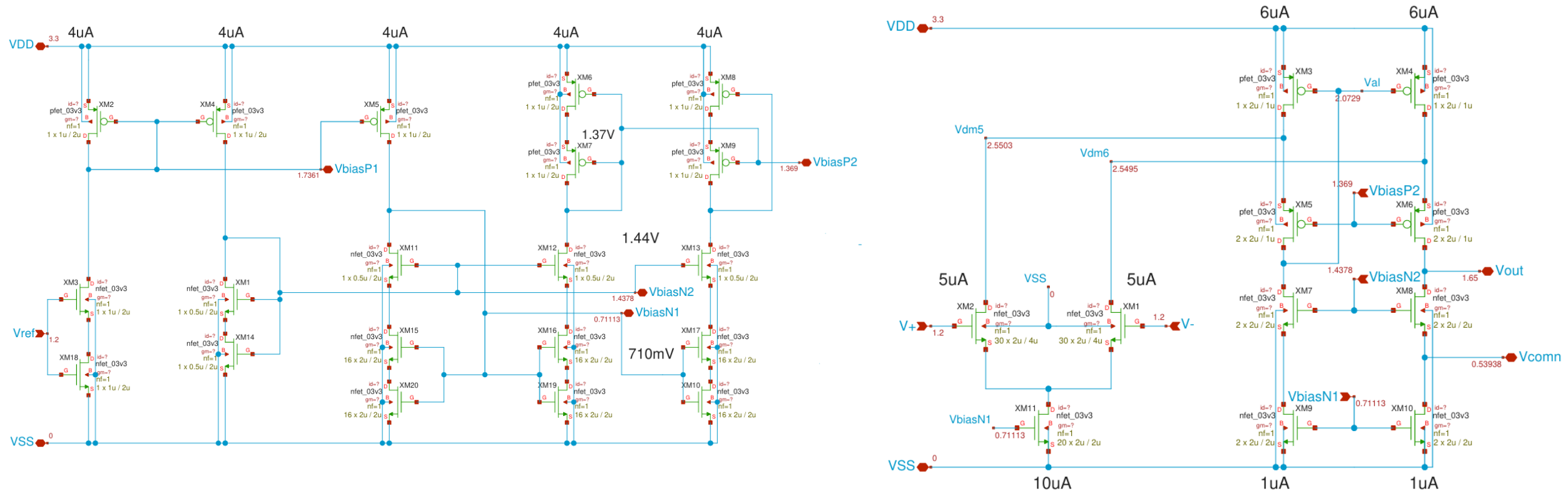


Schematic Flow

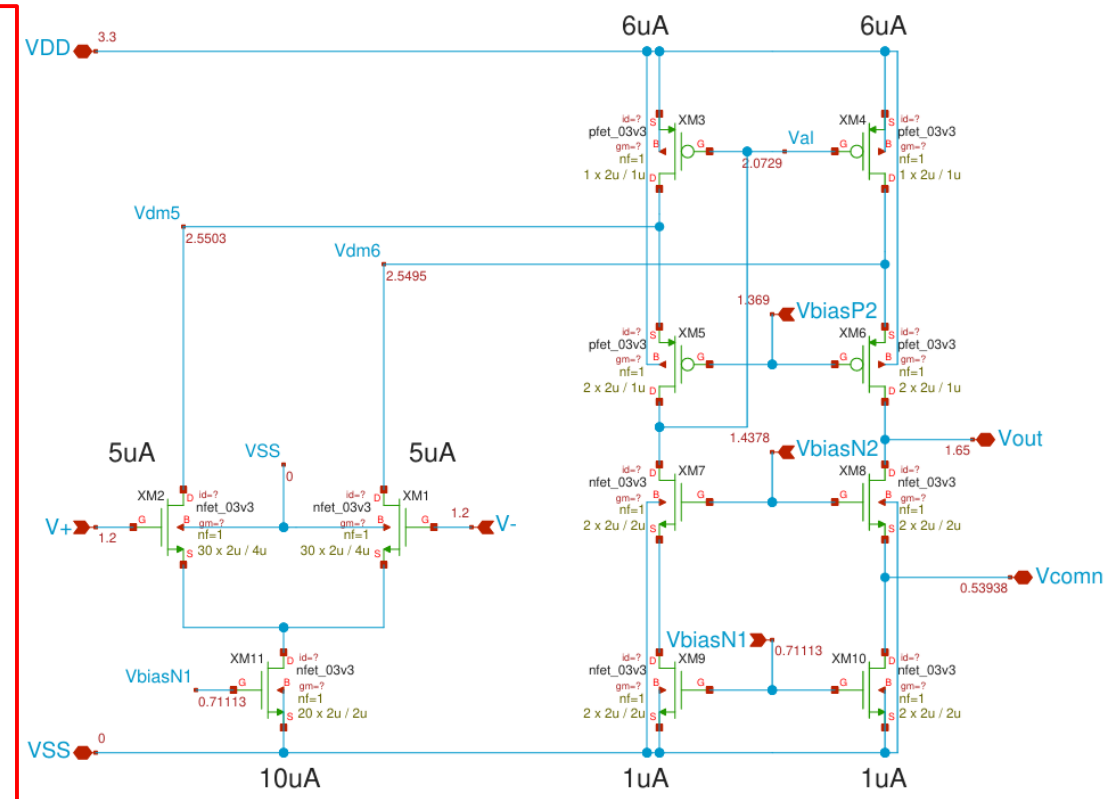
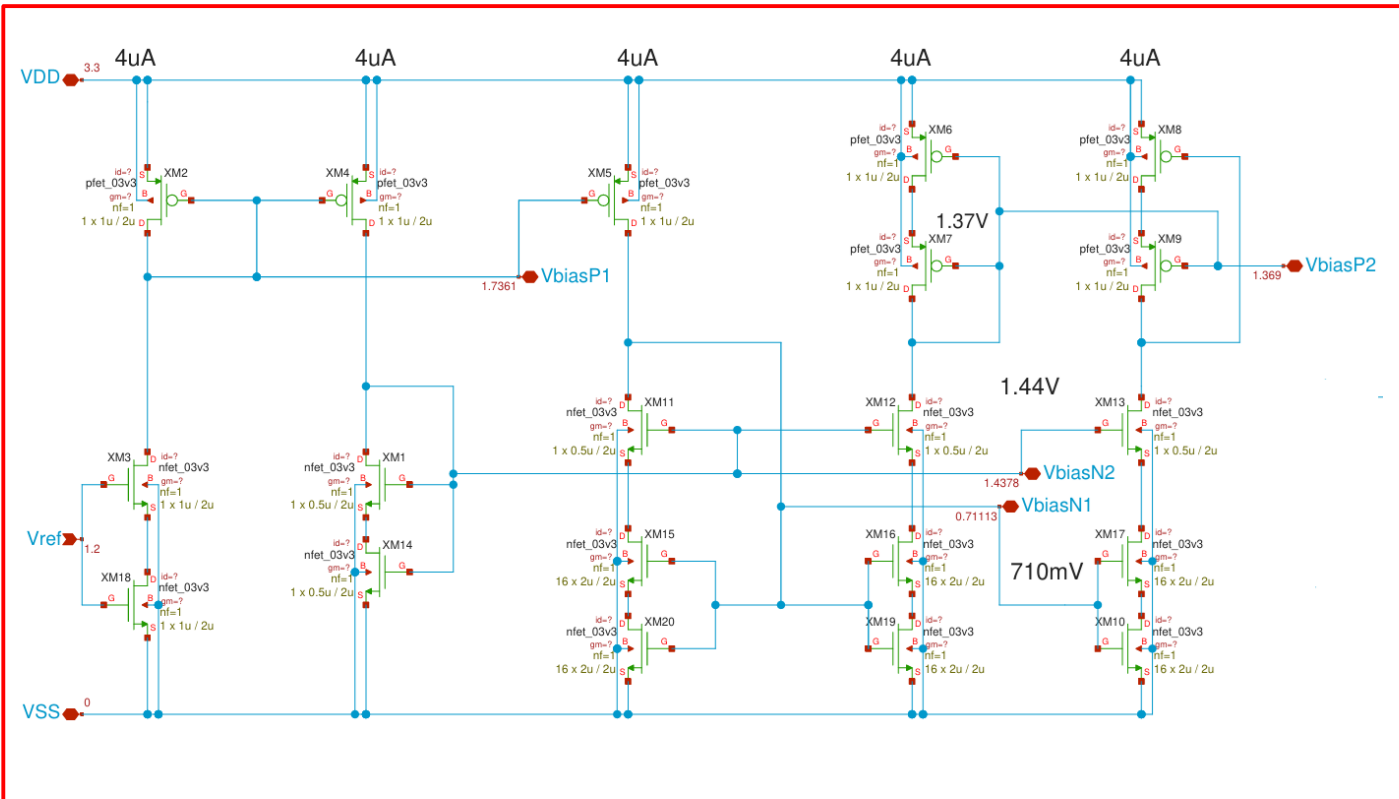
Current Reference Generator



Folded Cascode Amplifier

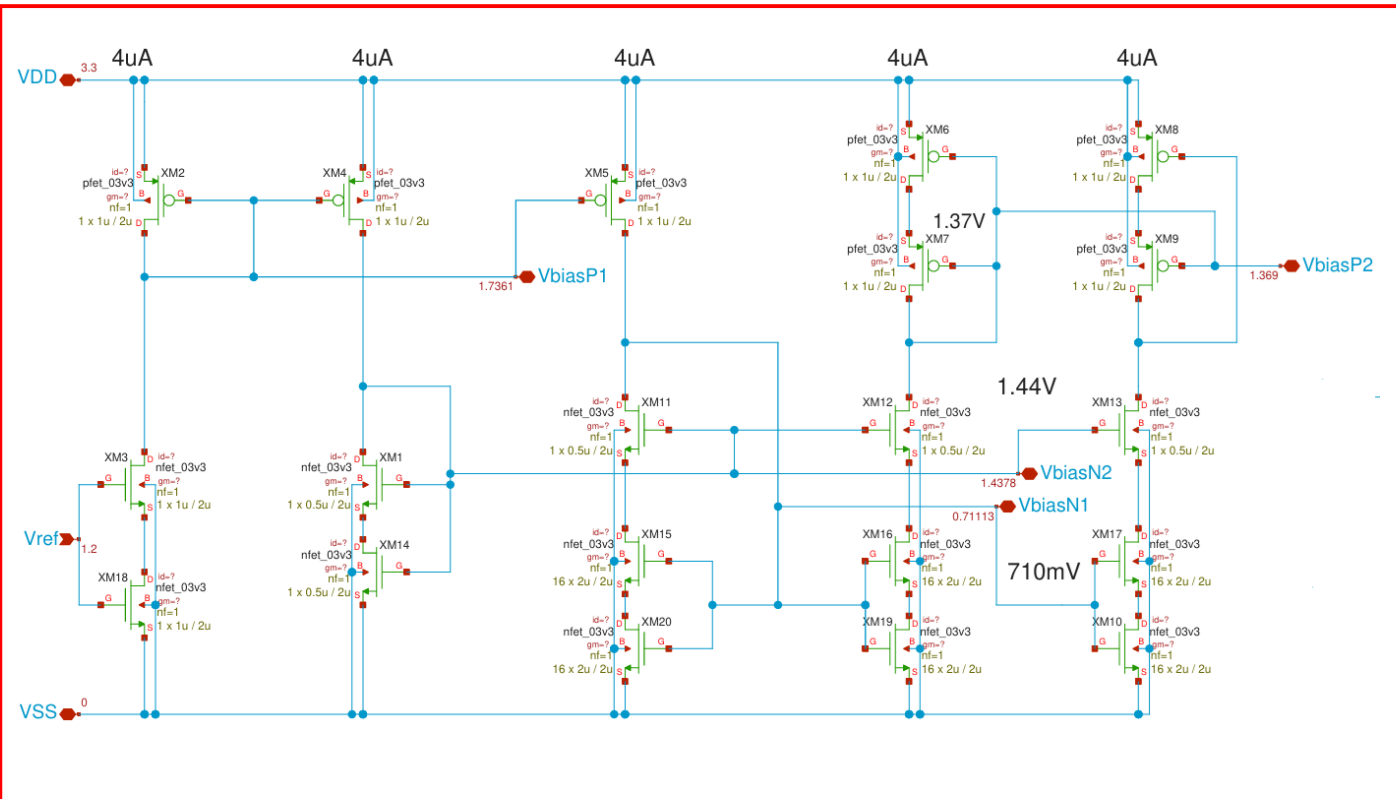


Folded Cascode Amplifier

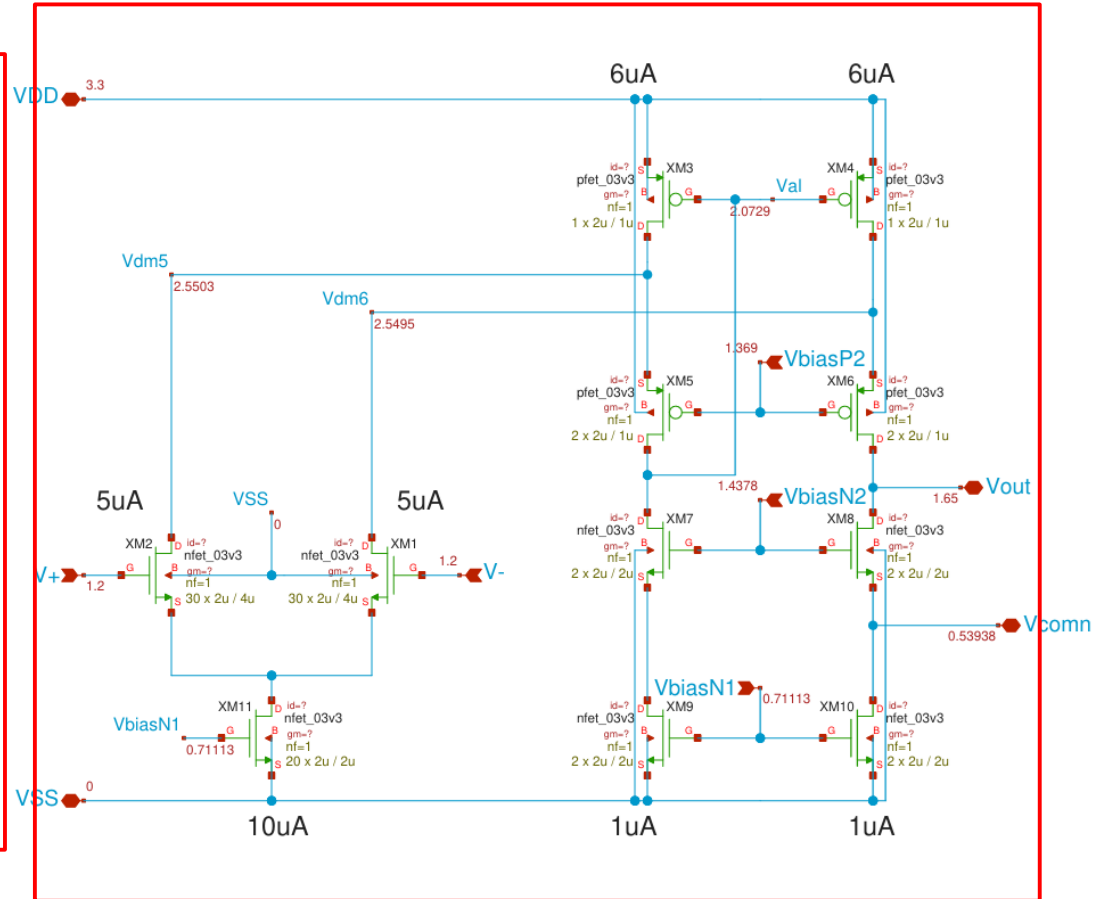


Bias circuit

Folded Cascode Amplifier

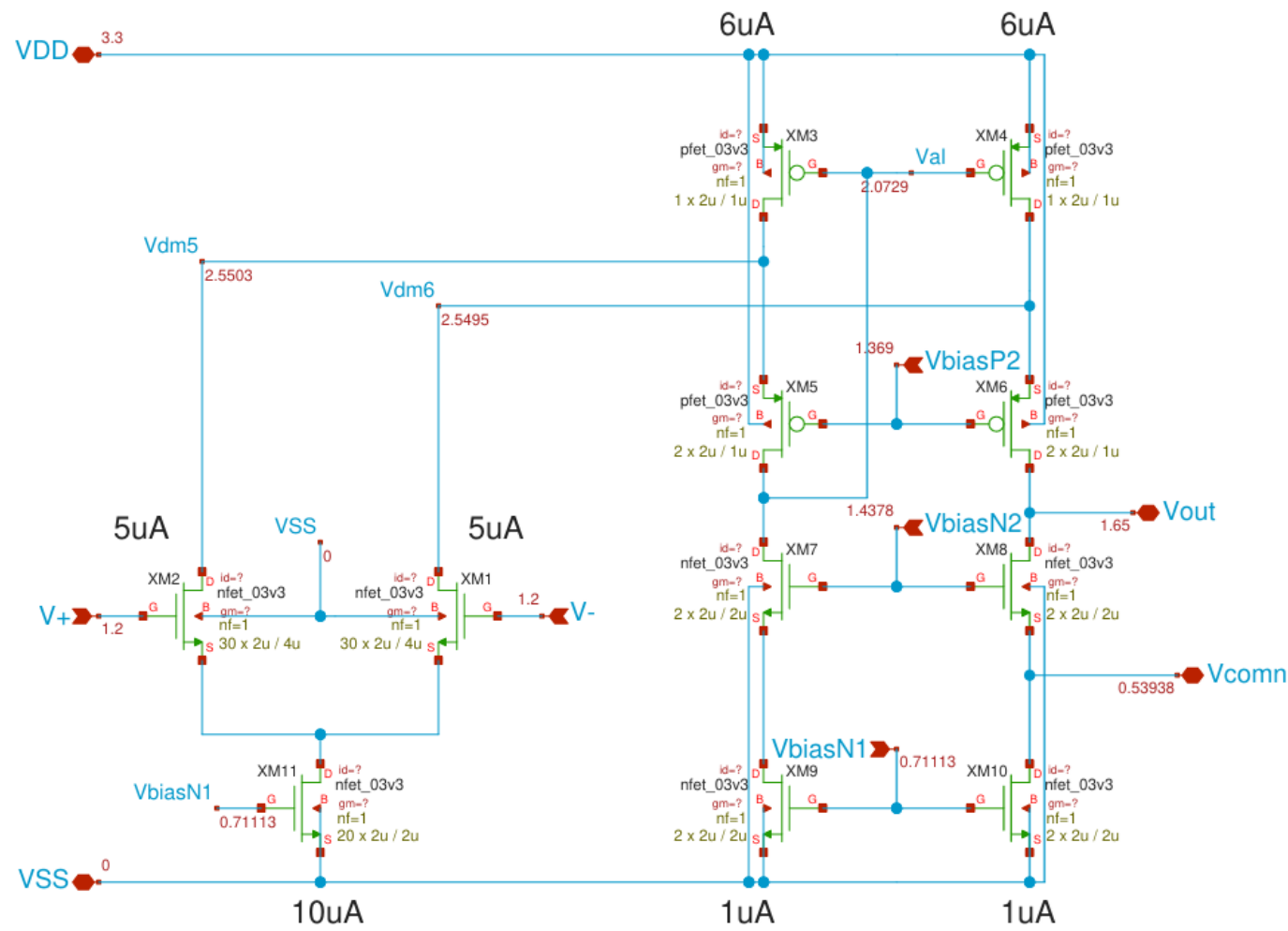


Bias circuit



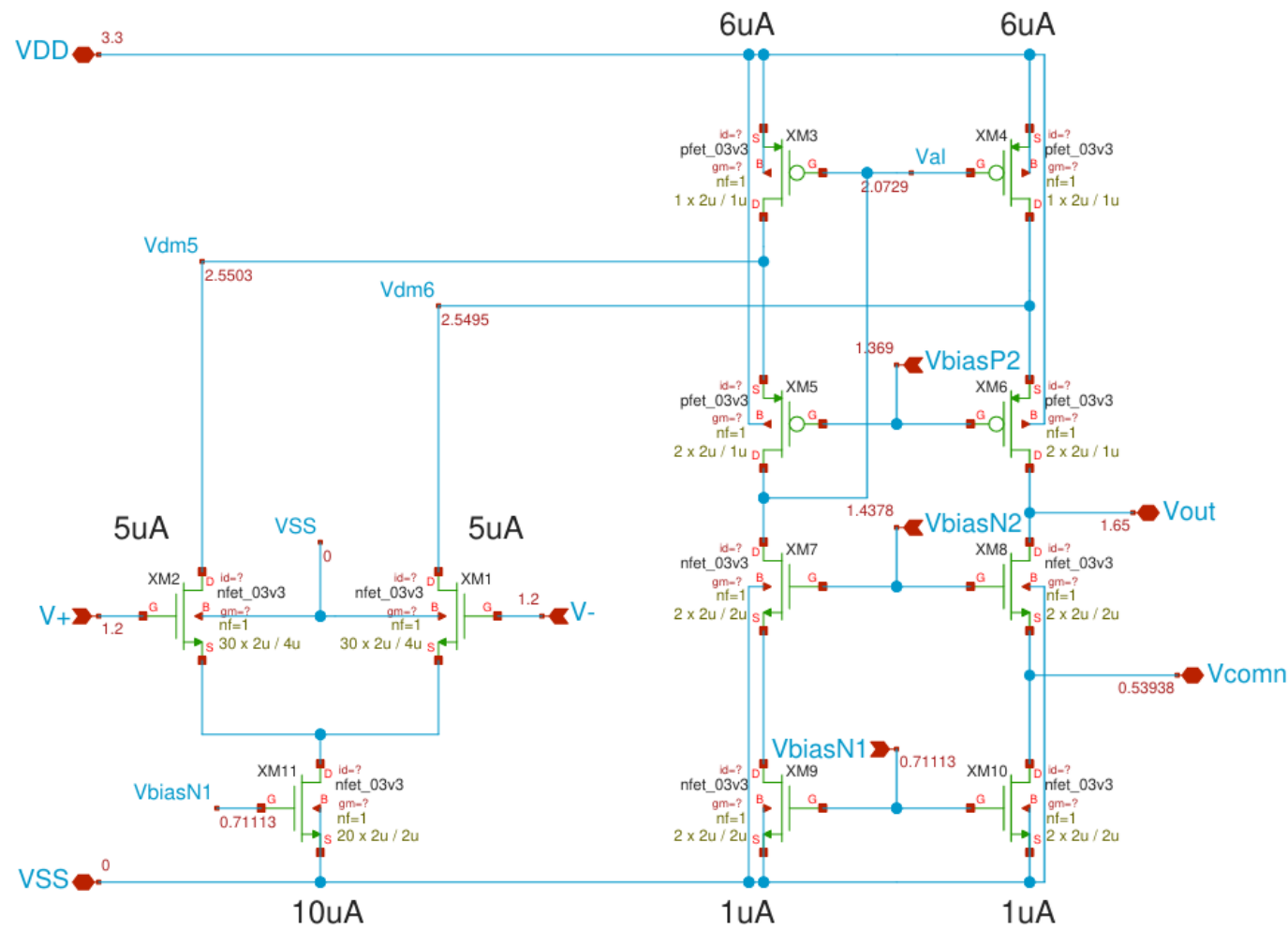
Core

Folded Cascode Amplifier - Aimed Specifications



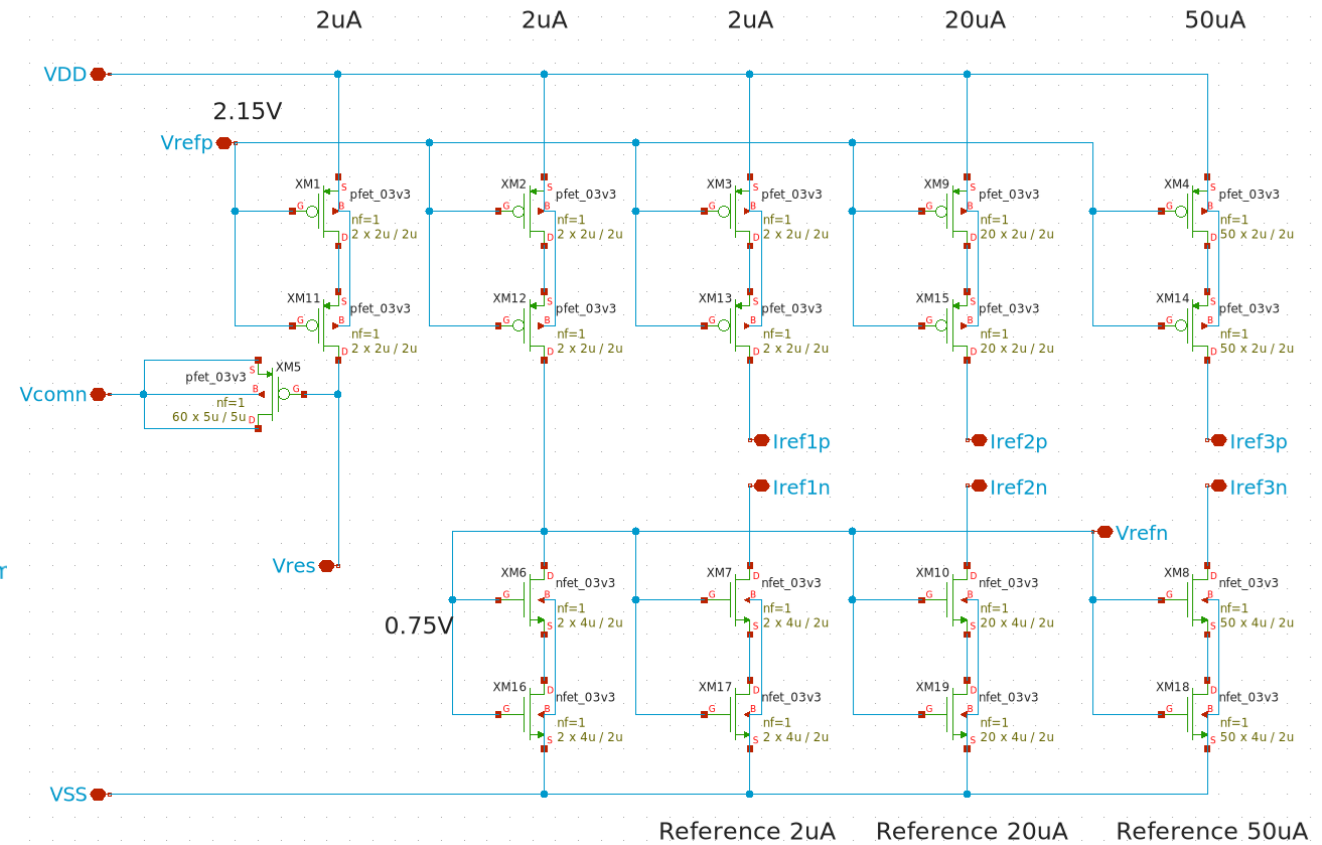
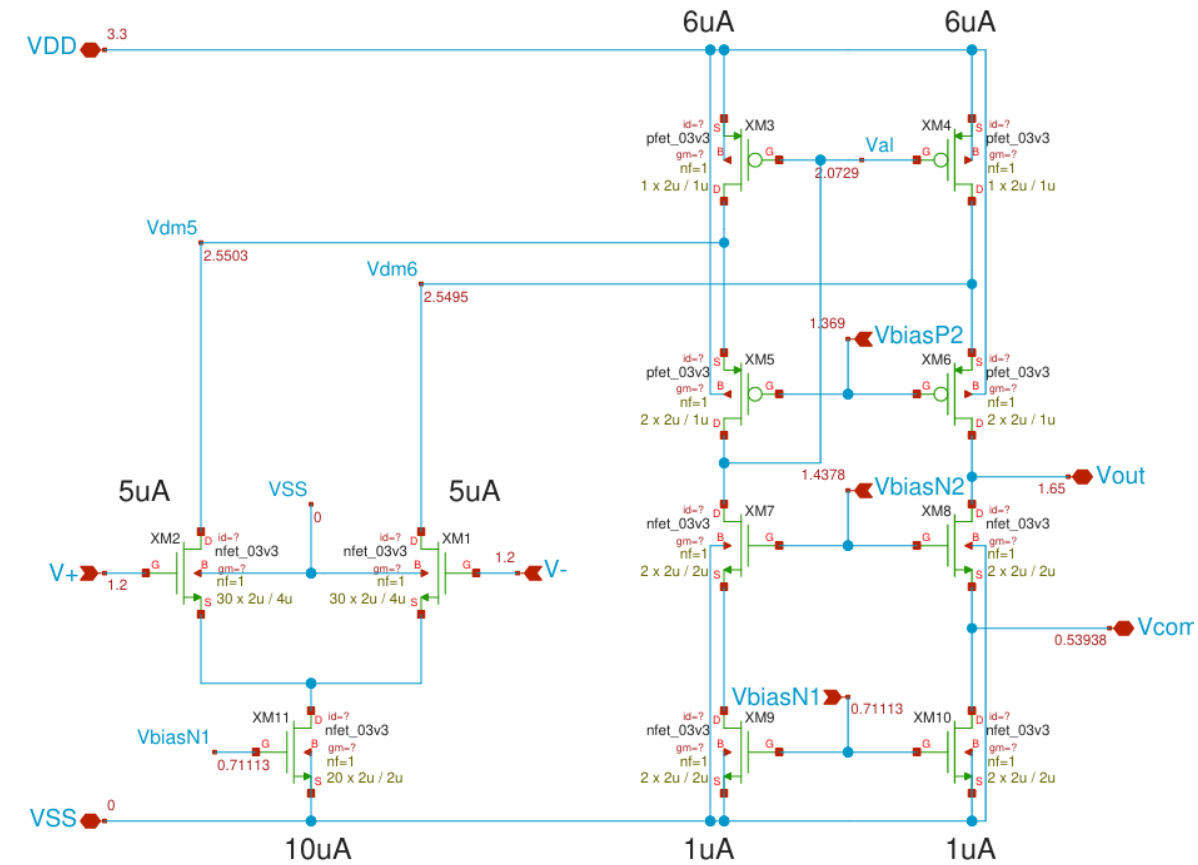
Specifications	Design Target		
	Min	Typ	Max
Voltage gain [dB]	90	-	-
Power [uW]	-	-	250
UGB [MHz]	1	-	-
Phase margin [°]	70	-	-
Input offset (3σ) [mV]	-	-	10

Folded Cascode Amplifier - post-layout results

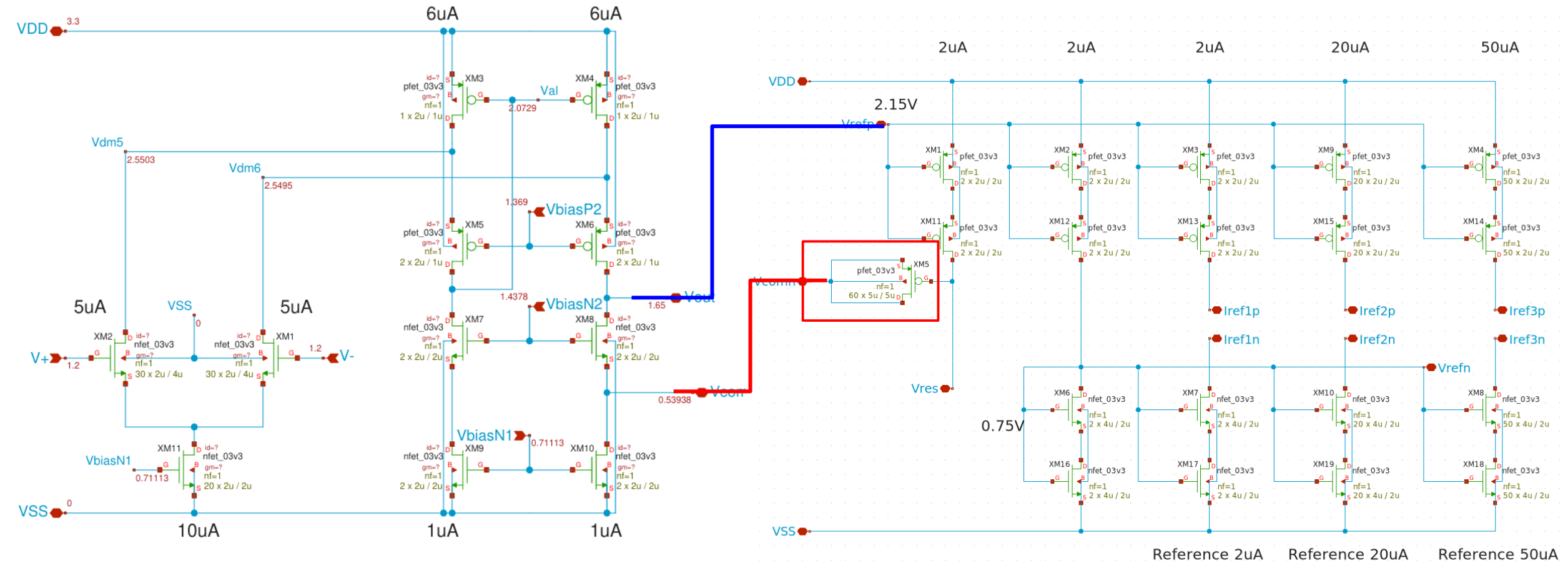


Specifications	Post-layout (CACE)		
	Min	Typ	Max
Voltage gain [dB]	91.5	103.7	104.9
Power [uW]	55.5	106.4	205
UGB [MHz]	1.4	2.6	4.7
Phase margin [°]	71.5	73.9	75.9
Input offset (3σ) [mV]	-	-	4.8

Second Stage: The Output Currents

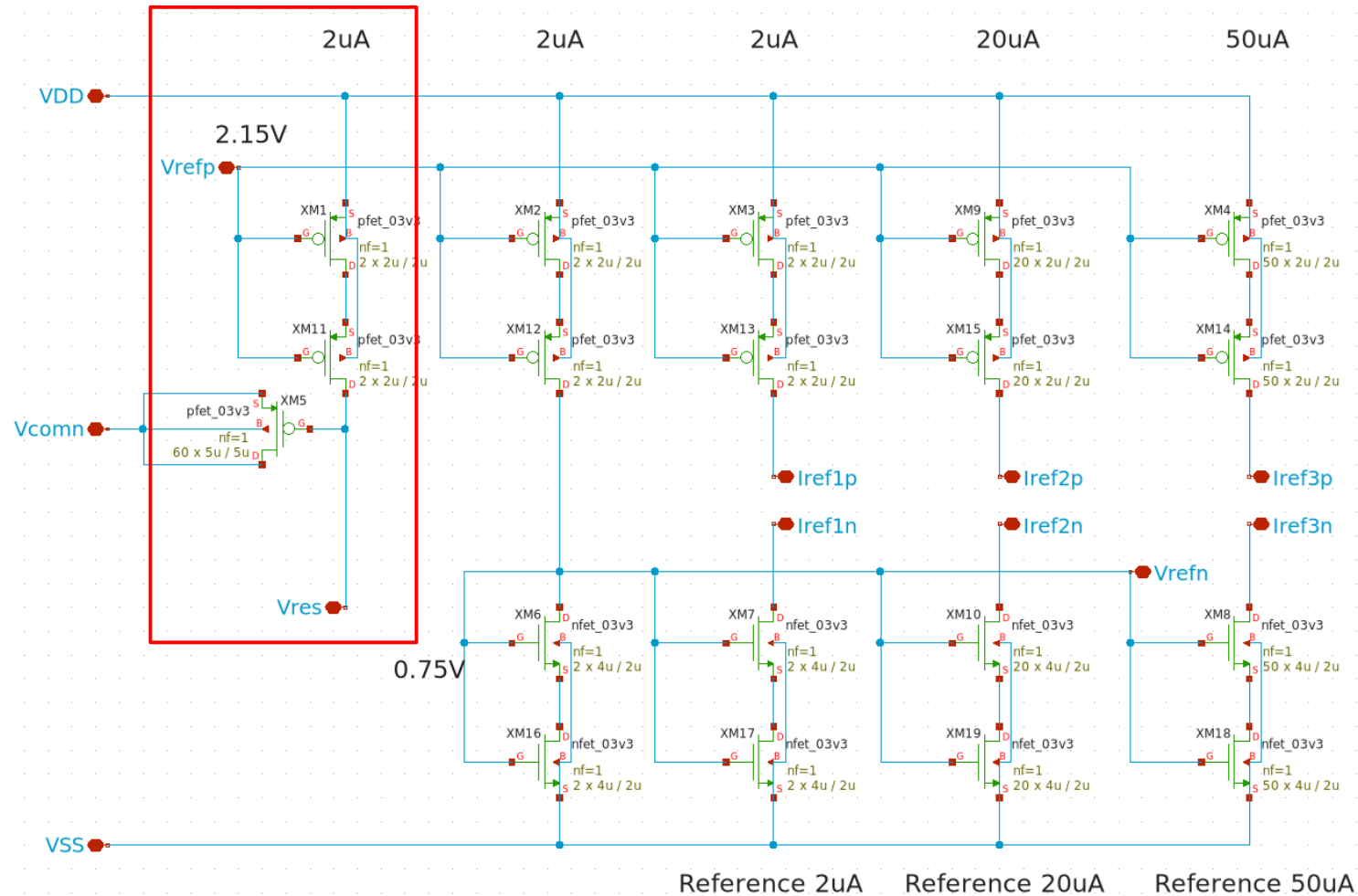


Second Stage: The Output Currents

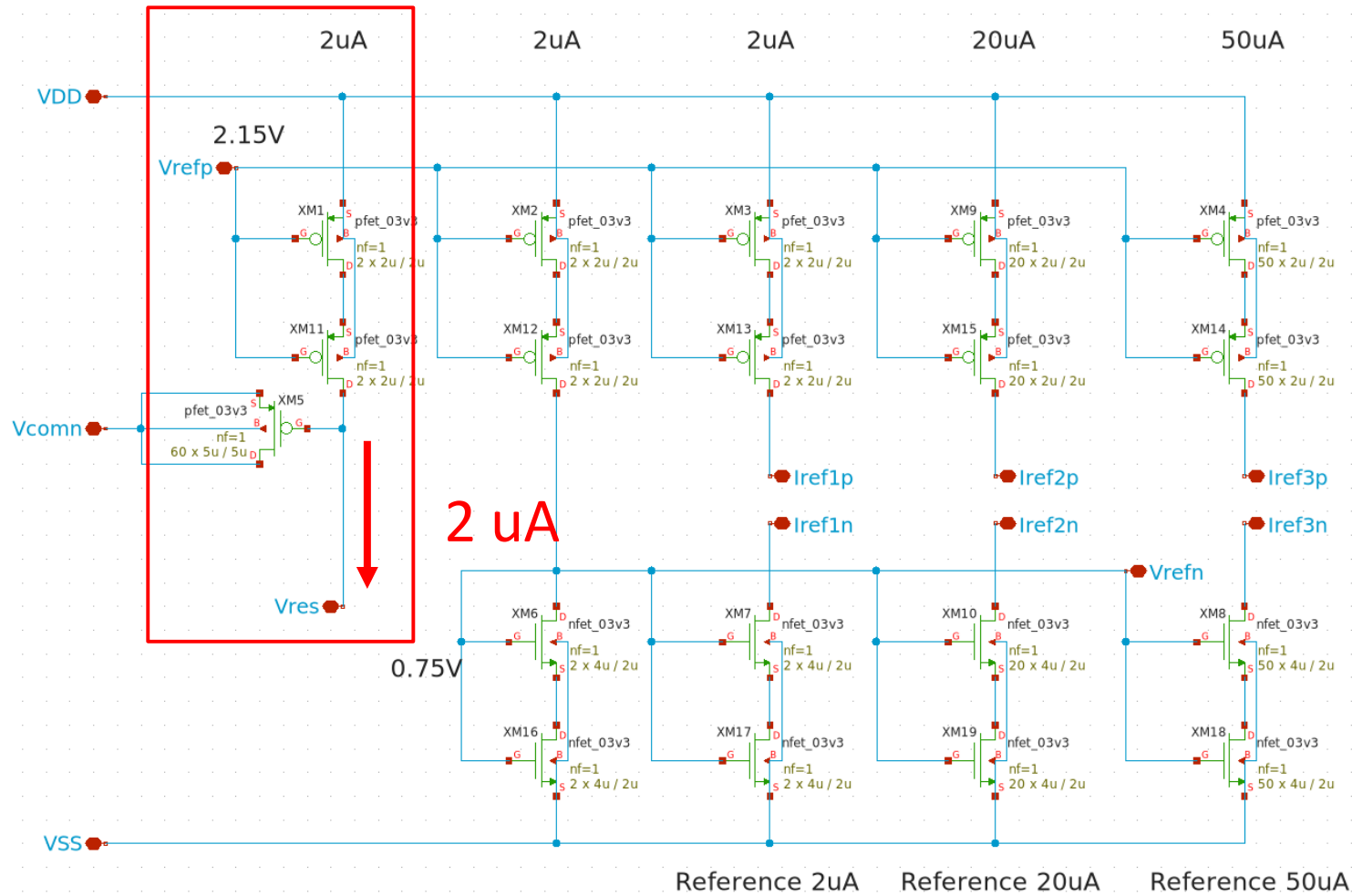


Nodes of the indirect compensation

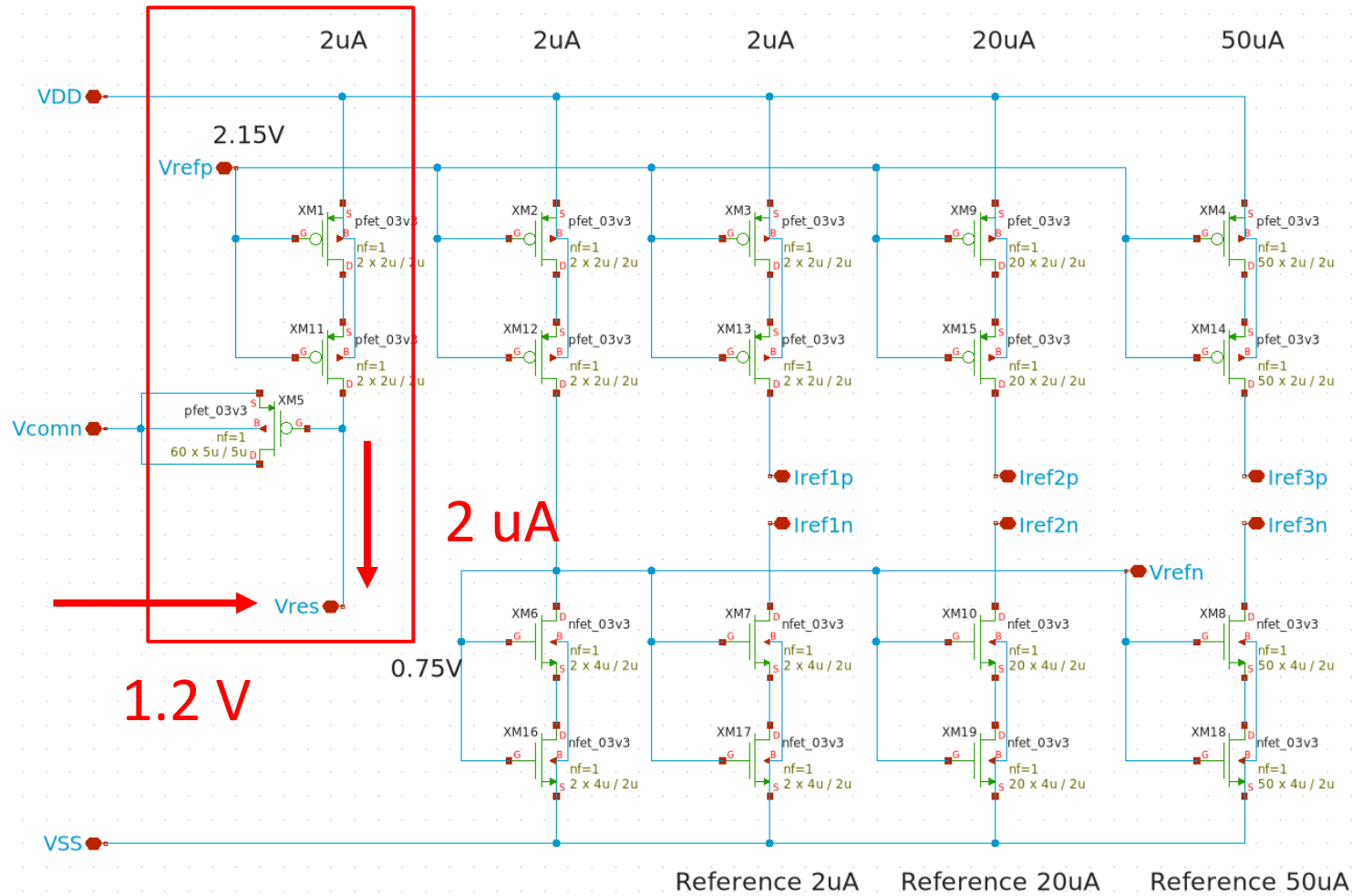
Second Stage - reference branch



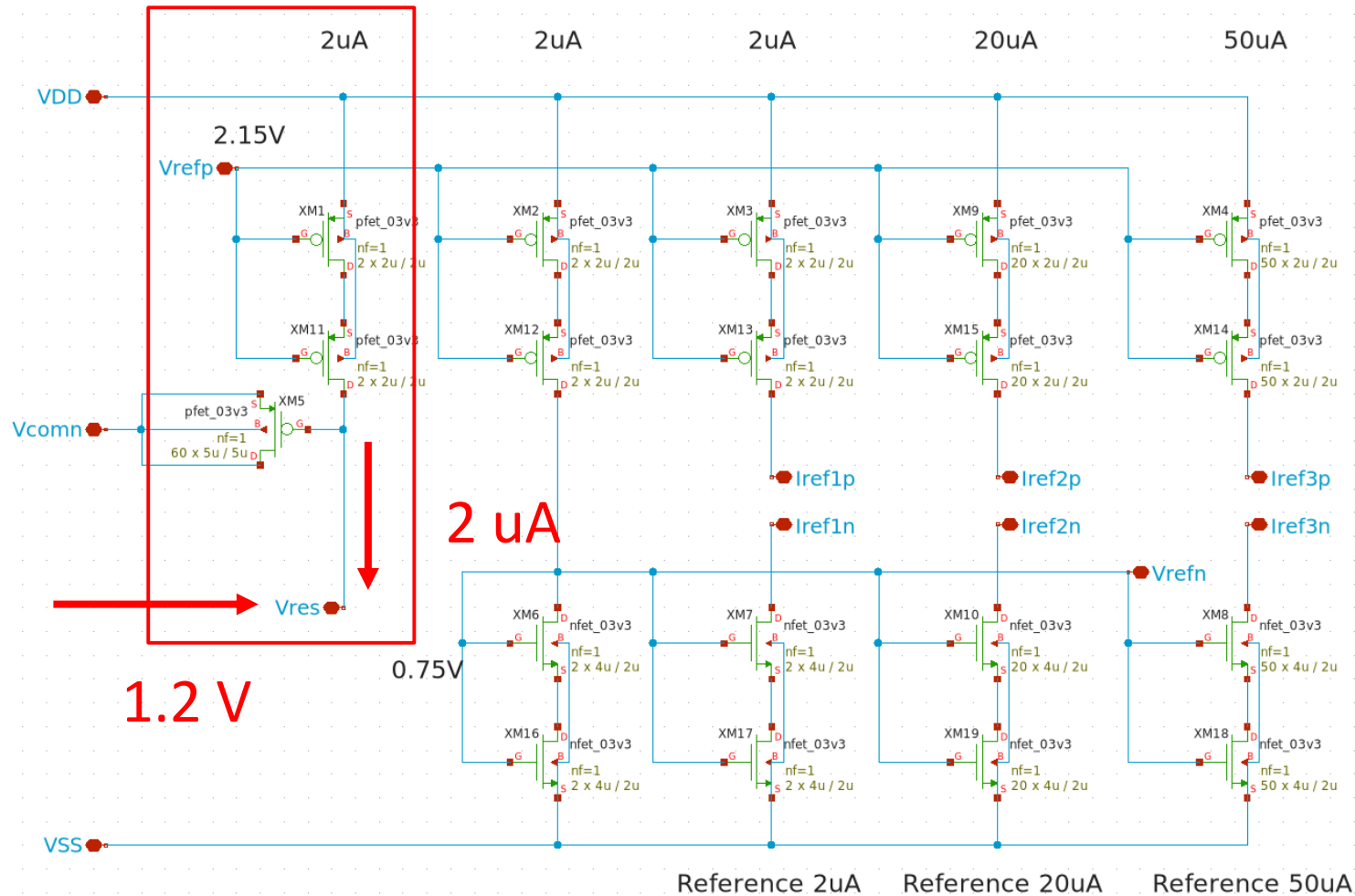
Second Stage - reference branch



Second Stage - reference branch



Second Stage - reference branch



$$R_{ext} = \frac{V_{ref} - V_{gnd}}{I_{ref}}$$

$$R_{ext} = \frac{1.2V}{2\mu A} = 600k\Omega$$

CACE Validation

CACE Structure

```
68 > default conditions:
69 >   VDD:
70 >     description: Analog power supply voltage
71 >     display: VDD
72 >     unit: V
73 >     typical: 3.3
74 >   Vref:
75 >     description: Voltage reference from BandGAP
76 >     display: Vref
77 >     unit: V
78 >     typical: 1.2
79 >   corner:
80 >     description: Process corner
81 >     display: Corner
82 >     typical: typical
83 >   temperature:
84 >     description: Ambient temperature
85 >     display: Temp
86 >     unit: °C
87 >     typical: 27
88
89 > parameters:
90 >   mc_param: ...
184
185 >   dc_param: ...
288
289 >   ac_param: ...
355
356 >   psr_param: ...
437
438 >   magic_area: ...
459
460 >   magic_drc: ...
470
471 >   netgen_lvs: ...
481
482 >   klayout_drc_feol: ...
492
493 >   klayout_drc_beol: ...
503
504 >   klayout_drc_full: ...
514
```

Project Structure

All directories, schematics, templates, testbenches, etc., of the main project are located in the directory `Onchip_CurRefGen/designs/libs/`. The project has the following hierarchical structure.

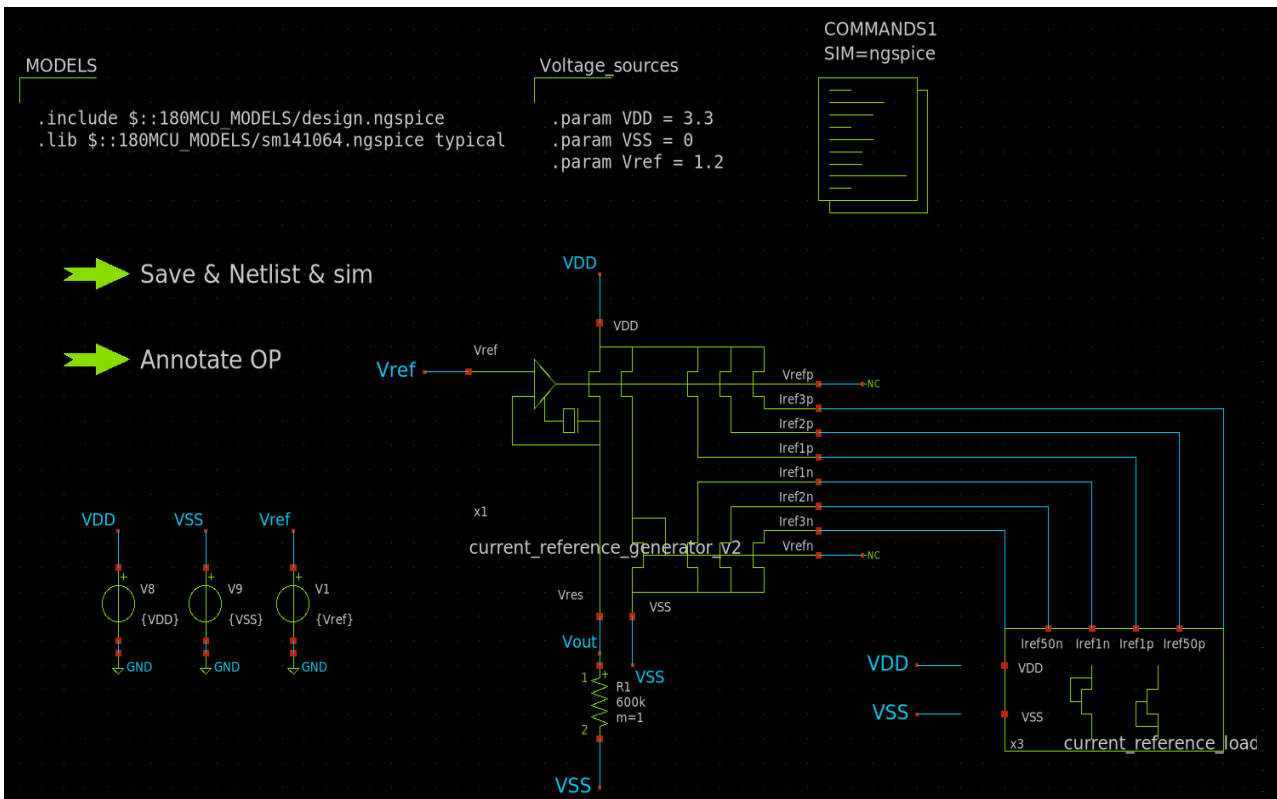
```
Current Reference Generator
|---Error Amplifier N_input
|   |---Error Amplifier N_input core
|   |---Error Amplifier N_input bias
|---Current Reference Second Stage
Current Reference Load
```

CACE Structure

Below is a brief description of a basic CACE structure to perform all the relevant measurements for the project and the use of the GF180 PDK. The complete CACE documentation can be found at the following [link](#).

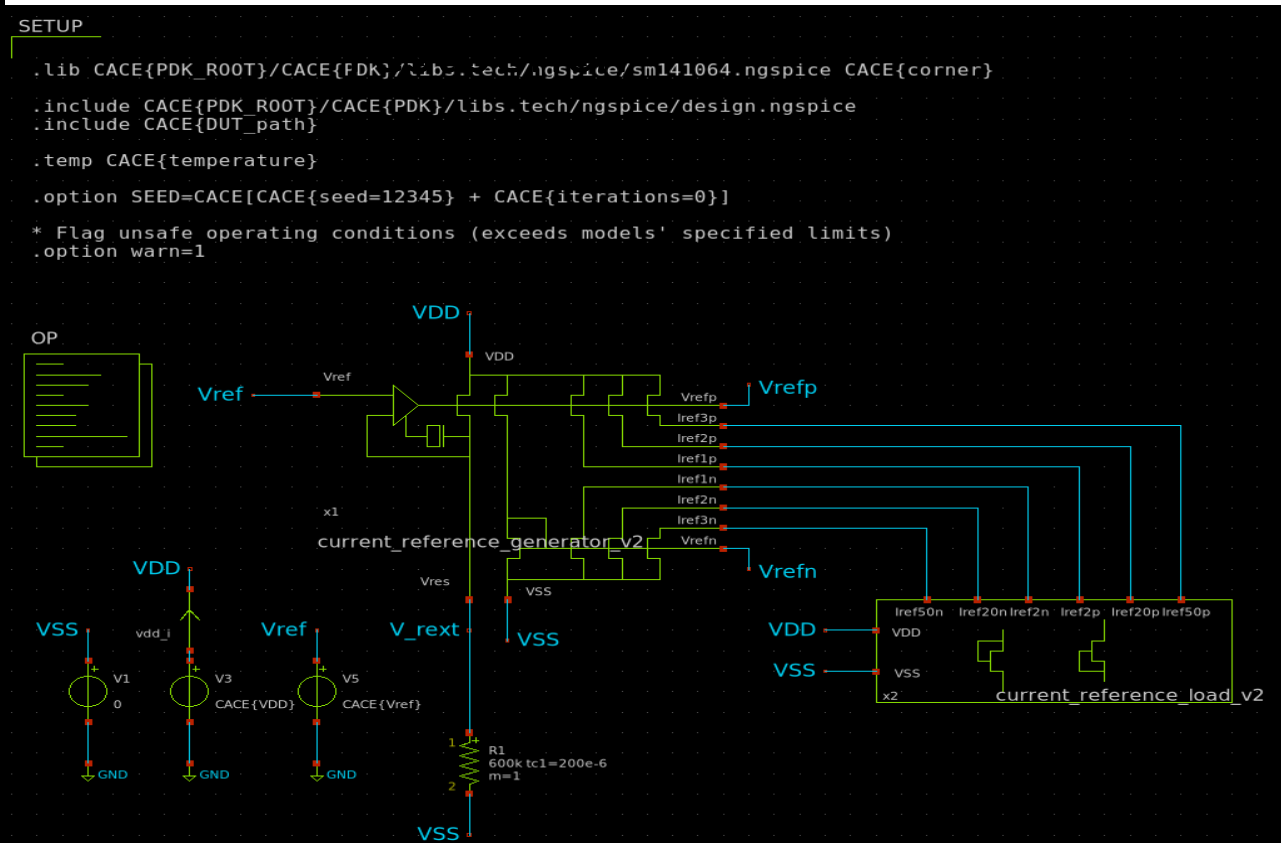
```
cace          # CACE datasheet (.yaml), templates, scripts
|---scripts   # Useful scripts for post-simulation processing, LVS, etc.
|---templates # Templates for running the parameters
gds           # Project layout and a script to flatten the layout
xschem        # Schematic, symbol designs, and all pre-layout testbenches
```

Schematic vs Template Testbenches



Schematic

Template Testbench



CACE Template Testbench

SETUP

```
.lib CACE{PDK_ROOT}/CACE{PDK}/libs.tech/ngspice/sml41064.ngspice CACE{corner}
.include CACE{PDK_ROOT}/CACE{PDK}/libs.tech/ngspice/design.ngspice
.include CACE{DUT_path}

.temp CACE{temperature}

.option SEED=CACE[CACE{seed=12345} + CACE{iterations=0}]

* Flag unsafe operating conditions (exceeds models' specified limits)
.option warn=1
```

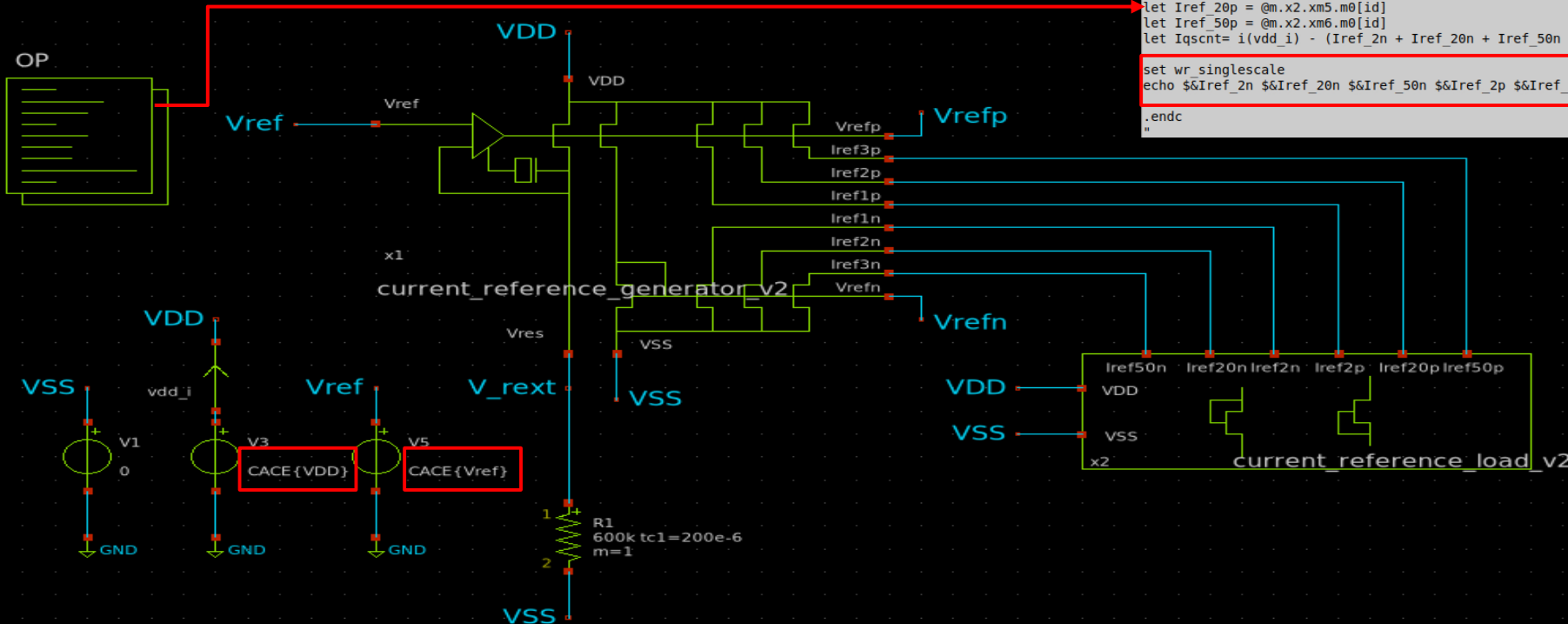
```
name=OP only_toplevel=true value="
.control
save all
```

```
*OP simulation
op
```

```
*TOP
```

```
let Iref_2n = @m.x2.xm1.m0[id]
let Iref_20n = @m.x2.xm2.m0[id]
let Iref_50n = @m.x2.xm3.m0[id]
let Iref_2p = @m.x2.xm4.m0[id]
let Iref_20p = @m.x2.xm5.m0[id]
let Iref_50p = @m.x2.xm6.m0[id]
let Iqscnt= i(vdd_i) - (Iref_2n + Iref_20n + Iref_50n + Iref_2p + Iref_20p + Iref_50p)
```

```
set wr_singlescale
echo $Iref_2n $Iref_20n $Iref_50n $Iref_2p $Iref_20p $Iref_50p $Iqscnt > CACE{simpath}/CACE{filename}_CACE{N}.data
.endc
"
```



Full Characterization (Operating Point & MonteCarlo)

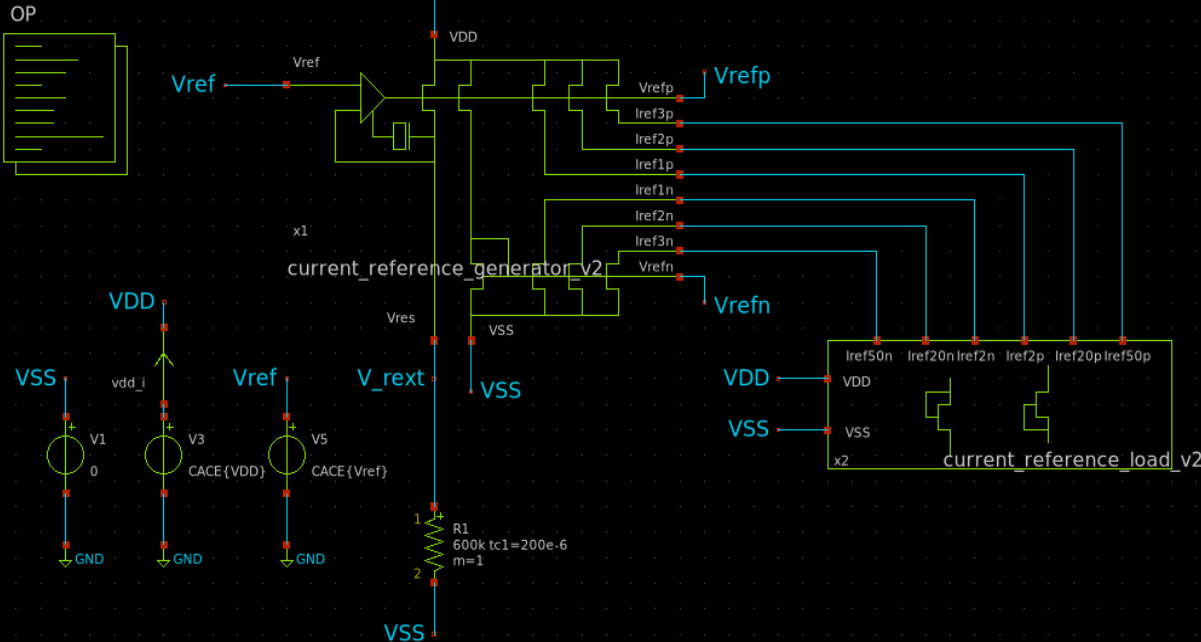
SETUP

```
.lib CACE{PDK_ROOT}/CACE{PDK}/libs.tech/ngspice/sm141064.ngspice CACE{corner}
.include CACE{PDK_ROOT}/CACE{PDK}/libs.tech/ngspice/design.ngspice
.include CACE{DUT_path}

.temp CACE{temperature}

.option SEED=CACE[CACE{seed=12345} + CACE{iterations=0}]

* Flag unsafe operating conditions (exceeds models' specified limits)
.option warn=1
```



tool:

ngspice:

```
template: current_reference_generator_v2_CACE_mc.sch
collate: iterations
format: ascii
suffix: .data
variables: [Iref_2n, Iref_20n, Iref_50n, Iref_2p, Iref_20p, Iref_50p]
script: current_reference_generator_mc_postprocess.py
script_variables: [Iref_2n_dict, Iref_20n_dict]
```

plot:

Iref_2n_arr:

```
type: histogram
axis: Iref_2n_dict
```

ref_20n_arr:

```
type: histogram
axis: Iref_20n_dict
```

conditions:

iterations: # for Monte Carlo

```
description: Iterations to run
display: Iterations
minimum: 1
maximum: 1000
step: linear
stepsize: 1
```

corner:

```
enumerate: [statistical]
```

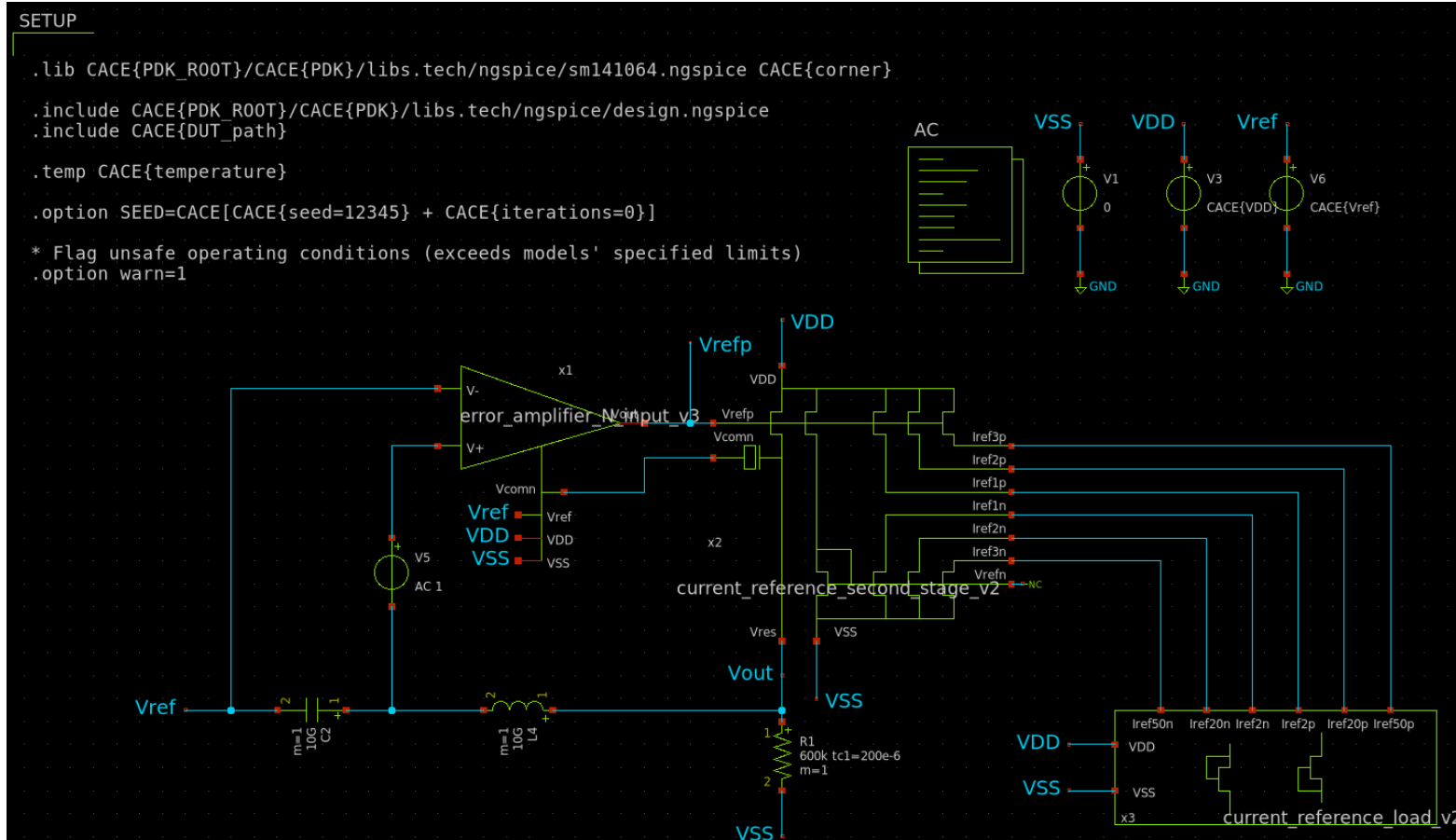
VDD:

```
typical: 3.3
```

temperature:

```
typical: 27
```

Full Characterization (AC)



```
tool:
  ngspice:
    template: current_reference_generator_v2_CACE_ac.sch
    format: ascii
    suffix: .data
    variables: [A0, UGB, PM, GM]

plot:
  gain_vs_temp:
    type: xyplot
    xaxis: temperature
    yaxis: A0

conditions:
  corner:
    enumerate: [ss, sf, fs, ff, typical]
  VDD:
    minimum: 2.97
    typical: 3.3
    maximum: 3.63
  temperature:
    minimum: -40
    typical: 27
    maximum: 125
```

Full Characterization (PSR)

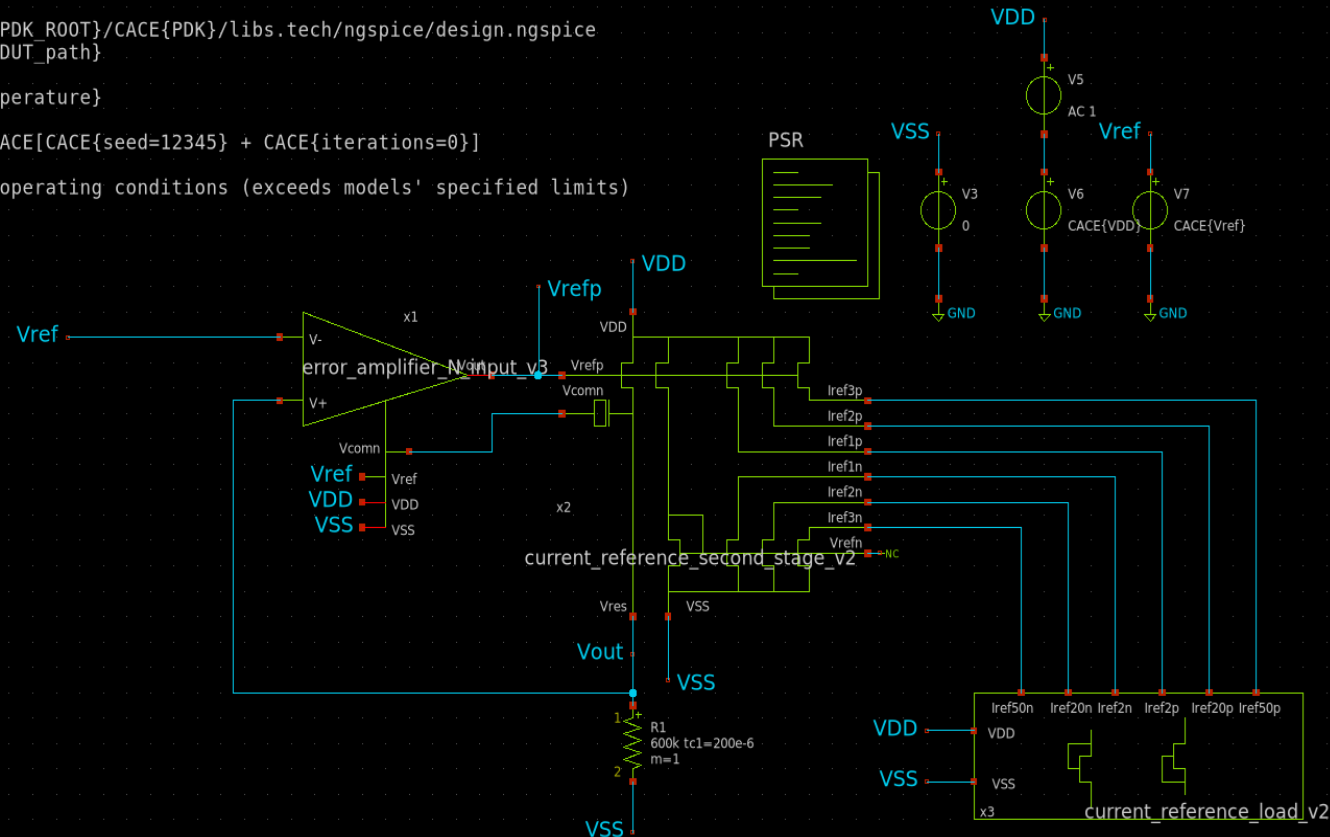
SETUP

```
.lib CACE{PDK_ROOT}/CACE{PDK}/libs.tech/ngspice/sm141064.ngspice CACE{corner}
.include CACE{PDK_ROOT}/CACE{PDK}/libs.tech/ngspice/design.ngspice
.include CACE{DUT_path}

.temp CACE{temperature}

.option SEED=CACE[CACE{seed=12345} + CACE{iterations=0}]

* Flag unsafe operating conditions (exceeds models' specified limits)
.option warn=1
```



tool:

ngspice:

```
template: current_reference_generator_v2_CACE_psr.sch
format: ascii
suffix: .data
variables: [psr_2n, psr_20n, psr_50n, psr_2p, psr_20p, psr_50p]
```

conditions:

corner:

```
enumerate: [ss, sf, fs, ff, typical]
```

VDD:

```
minimum: 2.97
typical: 3.3
maximum: 3.63
```

temperature:

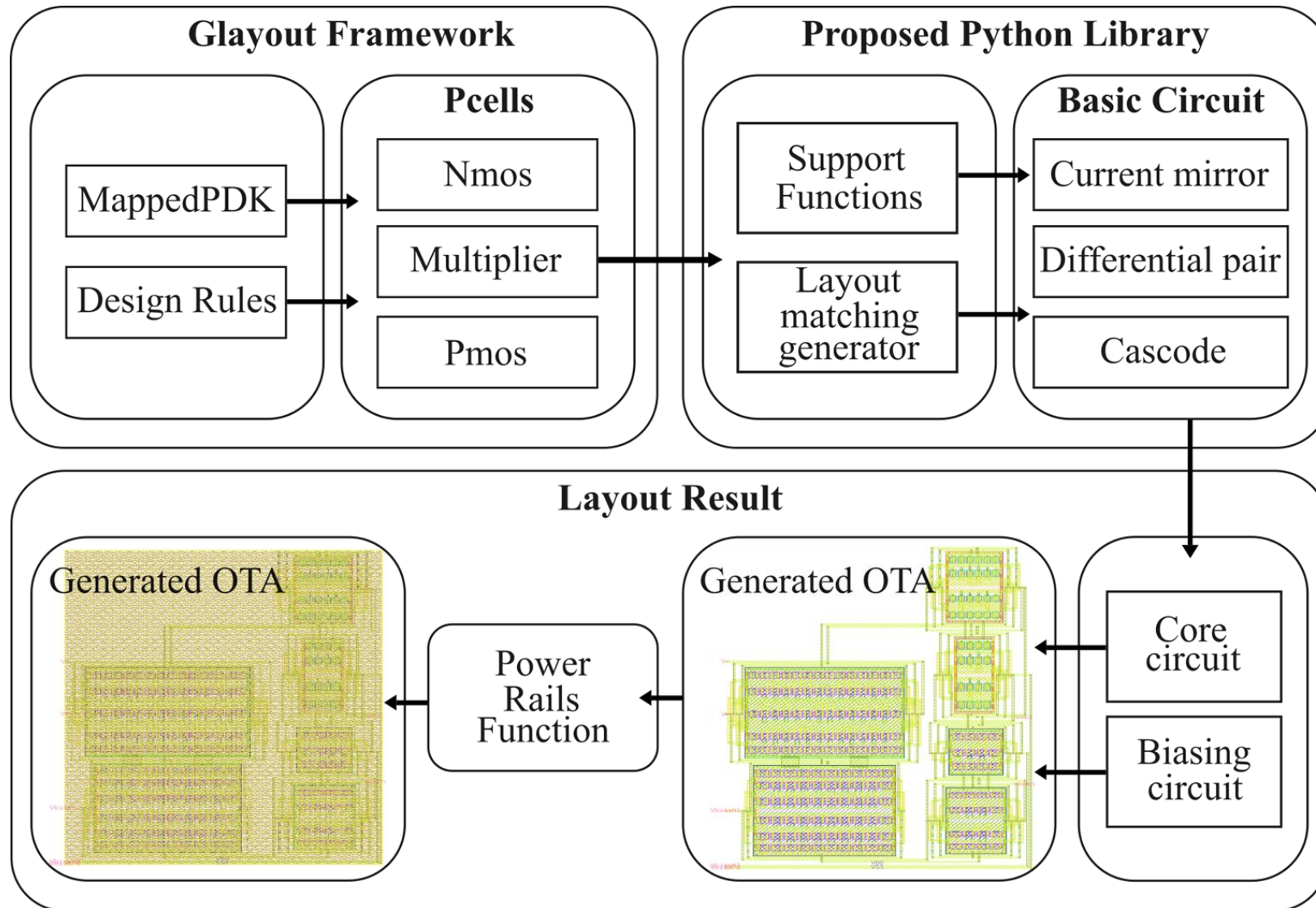
```
minimum: -40
typical: 27
maximum: 125
```

Results

Parameter	Tool	Result	Min Limit	Min Value	Typ Target	Typ Value	Max Limit	Max Value	Status
Iref 2n mc	ngspice	Iref_2n	any	1.839 uA	2 uA	2.049 uA	any	2.219 uA	Pass
Iref 20n mc	ngspice	Iref_20n	any	18.700 uA	20 uA	20.504 uA	any	22.384 uA	Pass
Iref 50n mc	ngspice	Iref_50n	any	47.253 uA	50 uA	51.303 uA	any	55.798 uA	Pass
Iref 2p mc	ngspice	Iref_2p	any	1.920 uA	2 uA	2.003 uA	any	2.095 uA	Pass
Iref 20p mc	ngspice	Iref_20p	any	19.124 uA	20 uA	20.038 uA	any	21.197 uA	Pass
Iref 50p mc	ngspice	Iref_50p	any	47.761 uA	50 uA	50.099 uA	any	52.501 uA	Pass
Iref 2n	ngspice	Iref_2n	any	1.989 uA	2 uA	2.049 uA	any	2.122 uA	Pass
Iref 20n	ngspice	Iref_20n	any	19.886 uA	20 uA	20.490 uA	any	21.217 uA	Pass
Iref 50n	ngspice	Iref_50n	any	49.716 uA	50 uA	51.224 uA	any	53.043 uA	Pass
Iref 2p	ngspice	Iref_2p	any	1.964 uA	2 uA	2.003 uA	any	2.031 uA	Pass
Iref 20p	ngspice	Iref_20p	any	19.641 uA	20 uA	20.030 uA	any	20.311 uA	Pass
Iref 50p	ngspice	Iref_50p	any	49.103 uA	50 uA	50.076 uA	any	50.776 uA	Pass
Iqscnt	ngspice	Iqscnt	any	84.594 uA	any	141.276 uA	any	238.778 uA	Pass
DC gain	ngspice	A0	any	56.453 dB	any	95.488 dB	any	104.182 dB	Pass
Unity Gain Bandwidth	ngspice	UGB	any	6.353 MHz	any	8.031 MHz	any	9.949 MHz	Pass
Phase Margin	ngspice	PM	45 °	75.065 °	any	78.627 °	any	81.845 °	Pass
Gain Margin	ngspice	GM	any	23.162 dB	any	23.673 dB	any	24.345 dB	Pass
PSR Iref 2n	ngspice	psr_2n	any	-114.029 dB	any	-113.769 dB	any	-113.466 dB	Pass
PSR Iref 20n	ngspice	psr_20n	any	-94.029 dB	any	-93.769 dB	any	-93.466 dB	Pass
PSR Iref 50n	ngspice	psr_50n	any	-86.070 dB	any	-85.810 dB	any	-85.507 dB	Pass
PSR Iref 2p	ngspice	psr_2p	any	-114.137 dB	any	-113.966 dB	any	-113.846 dB	Pass
PSR Iref 20p	ngspice	psr_20p	any	-94.137 dB	any	-93.966 dB	any	-93.846 dB	Pass
PSR Iref 50p	ngspice	psr_50p	any	-86.178 dB	any	-86.007 dB	any	-85.887 dB	Pass

Automated Layout

Layout Hierarchy Flow



Automatic placement with layout matching techniques

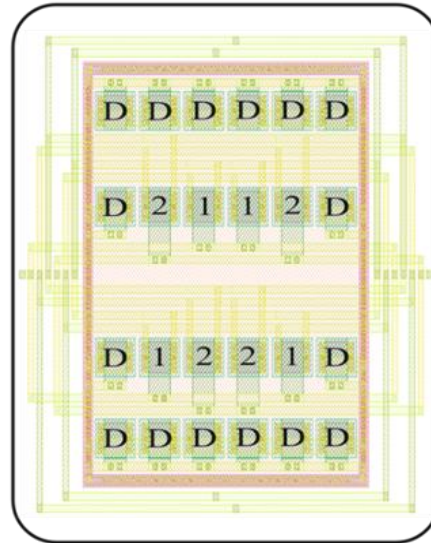
Input Matrix

$$\begin{bmatrix} 2 & 1 & 1 & 2 \\ 1 & 2 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 1 \end{bmatrix}$$

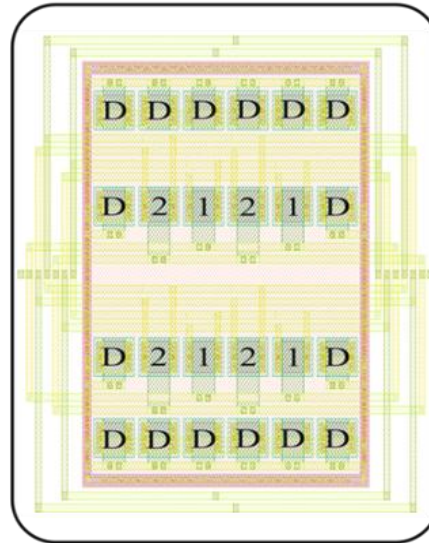
$$\begin{bmatrix} 3 & 2 & 1 & 1 & 2 & 3 \end{bmatrix}$$

Automated
Layout
Generated



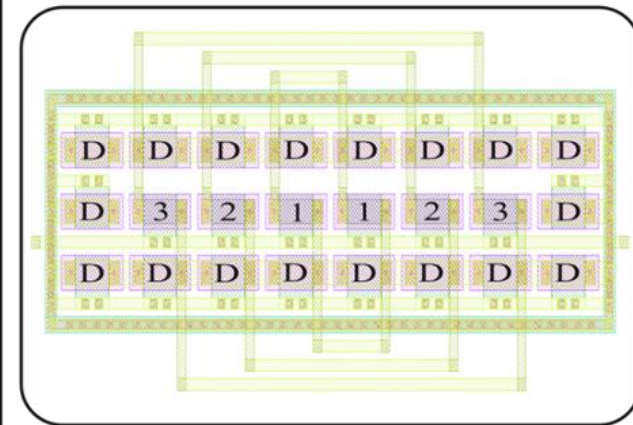
(a)

Common-Centroid
Two Transistors
Placement



(b)

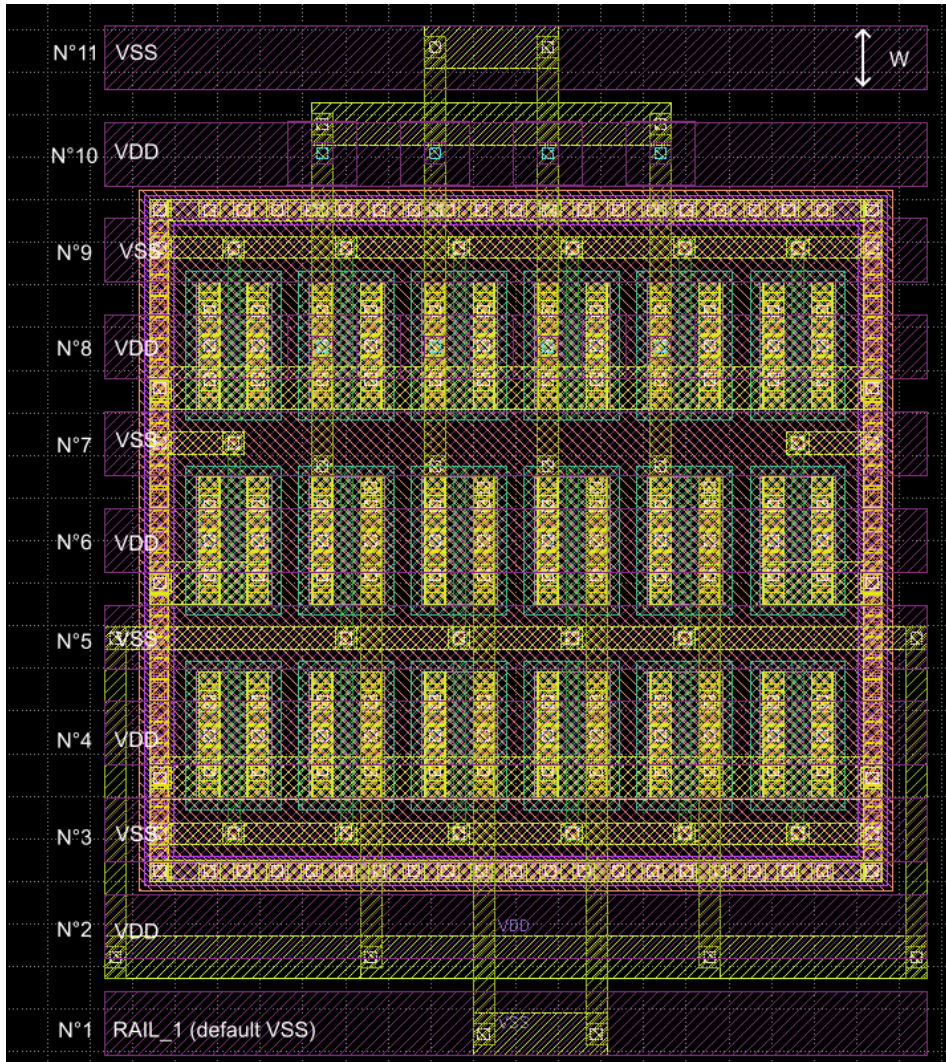
Interdigitated
Two Transistors
Placement



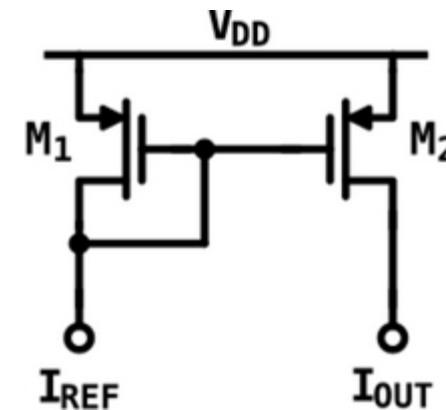
(c)

Interdigitated For
N-Transistors Placement

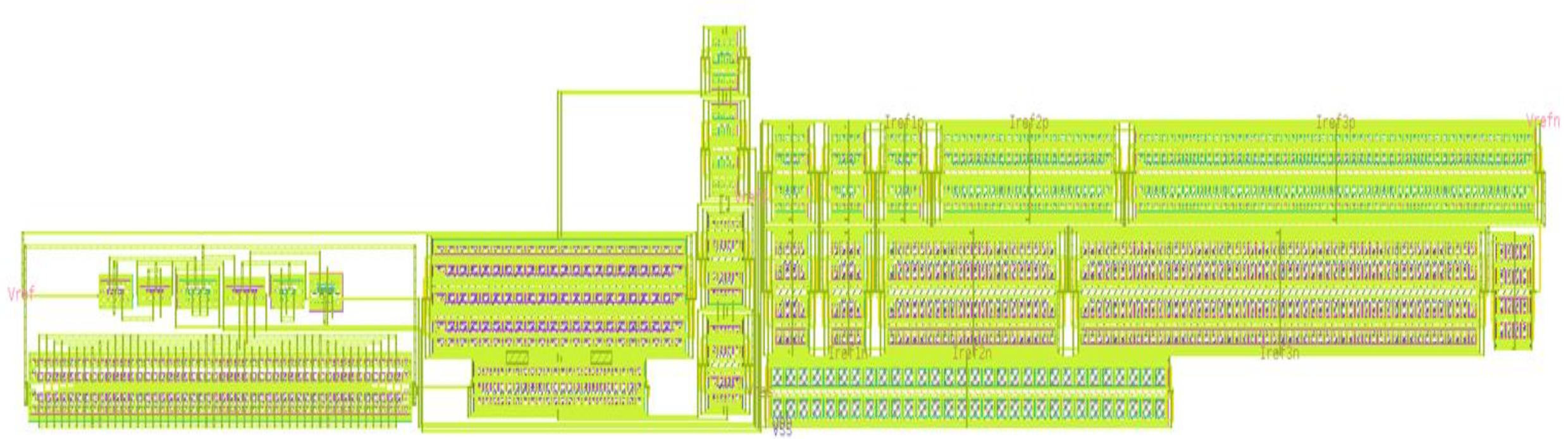
Power rails placement function



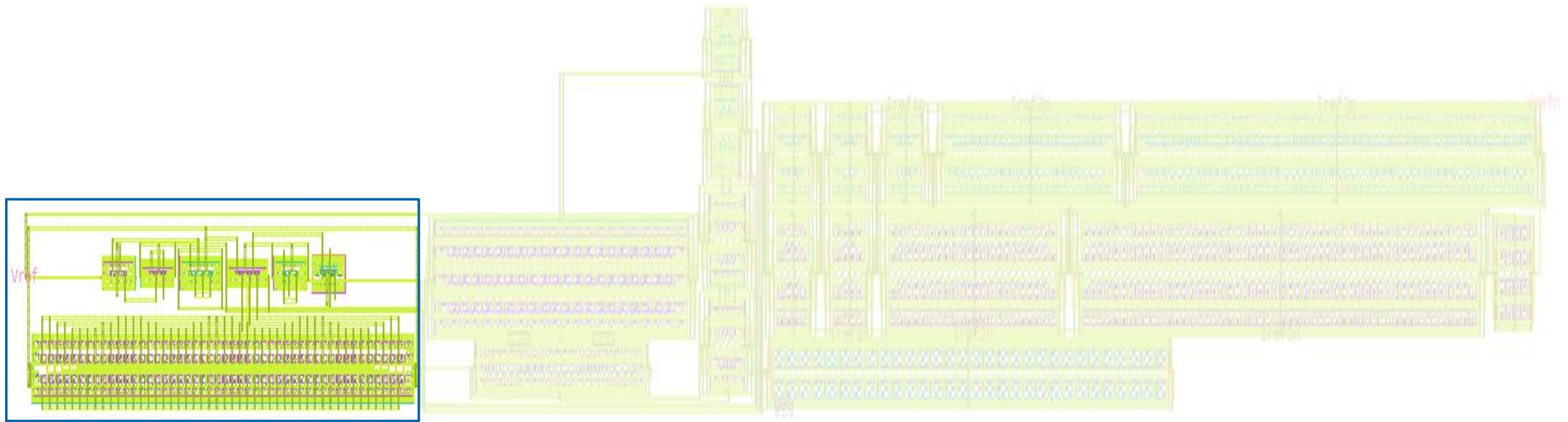
- ▶ W configurable.
- ▶ Configurable rails:
 - Default: VSS - VDD - ... - VDD - VSS.
 - Rail number and name.
- ▶ Adjusted to component size.
- ▶ Automatic connection to the indicated ports.



Current Reference layout generated

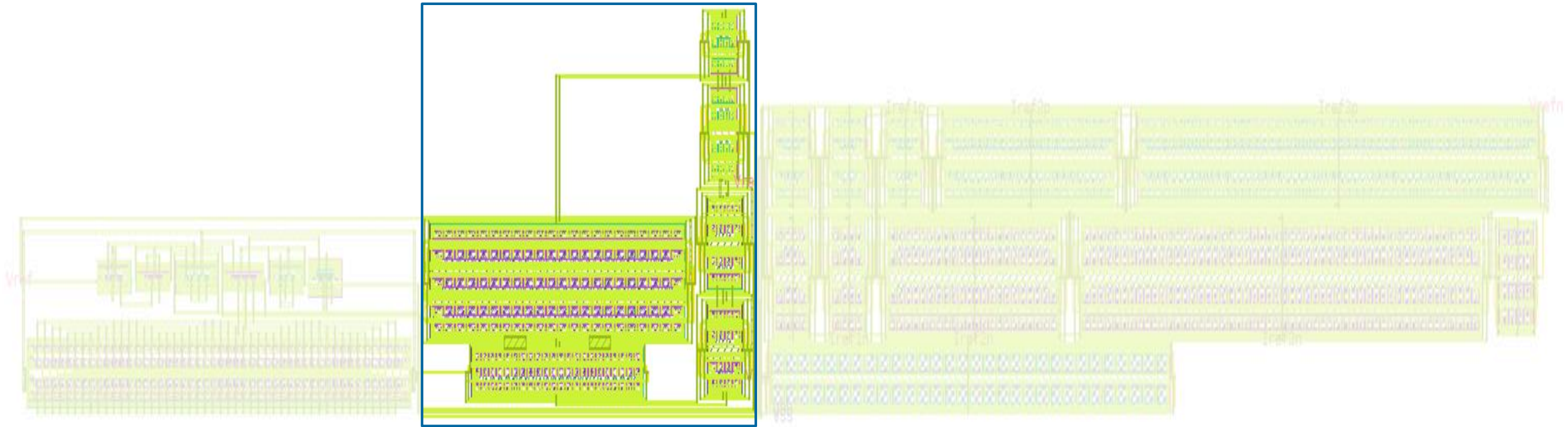


Current Reference layout generated



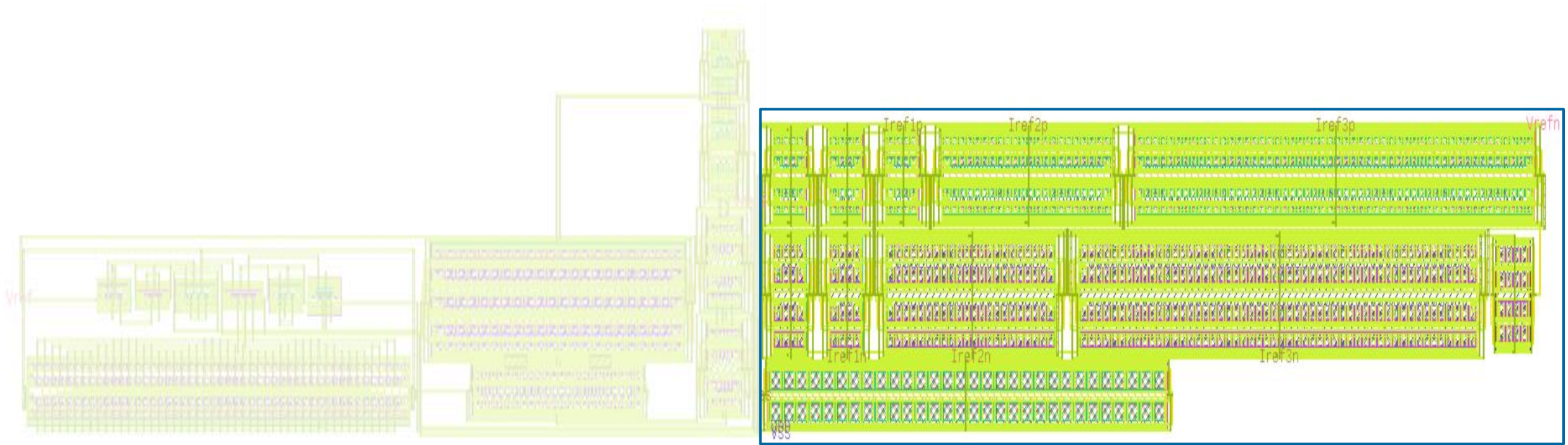
Bias Circuit

Current Reference layout generated



Folded Cascode
OTA Circuit

Current Reference layout generated



Current Outputs Circuit

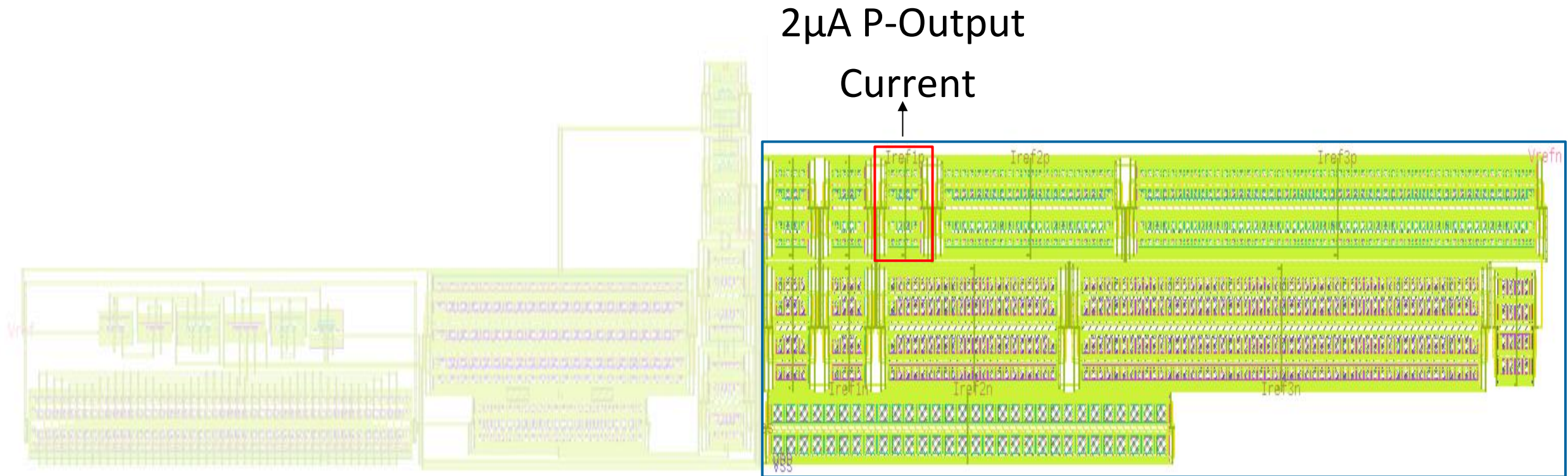
Current Reference layout generated

```
def generator_current_reference():  
    # Information of the transistors  
  
    m1 = {'type':'pfet', 'name':'M1', 'width':2, 'length':2, 'width_route_mult':1, 'fingers':1, 'with_substrate_tap':False, 'with_tie':True, 'with_dummy':True, 'lvt':True}  
    m2 = {'type':'pfet', 'name':'M2', 'width':2, 'length':2, 'width_route_mult':1, 'fingers':1, 'with_substrate_tap':False, 'with_tie':True, 'with_dummy':True, 'lvt':True}  
    m5 = {'type':'nfet', 'name':'M5', 'width':4, 'length':2, 'width_route_mult':1, 'fingers':1, 'with_substrate_tap':False, 'with_tie':True, 'with_dummy':True, 'lvt':False}  
    MC = {'type':'pfet', 'name':'MC', 'width':5, 'length':5, 'width_route_mult':1, 'fingers':1, 'with_substrate_tap':False, 'with_tie':True, 'with_dummy':False, 'lvt':False}  
    devices_info = [m1, m2, m5, MC]  
    matrix1 = [[1,2],[2,1]]  
    matrix2_p = [[1,2],[2,1]]  
    #For design convenience, matrix1 = matrix2_p to improve layout distribution  
    matrix3_n = [[1,2],[2,1]]  
    pmirror = [1,10,25]  
    nmirror = [1,10,25]  
    MC_array = [[1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1], [1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1]]  
  
    current outs, rails route list = current reference(sky130, devices info, matrix1, matrix2 p, matrix3 n, pmirror, nmirror, MC array, moscaps=True)
```

Current Outputs Function

- ▶ Current outputs configurables with input arguments from the function.
- ▶ 1, 10 and 25 times the 2uA current reference.

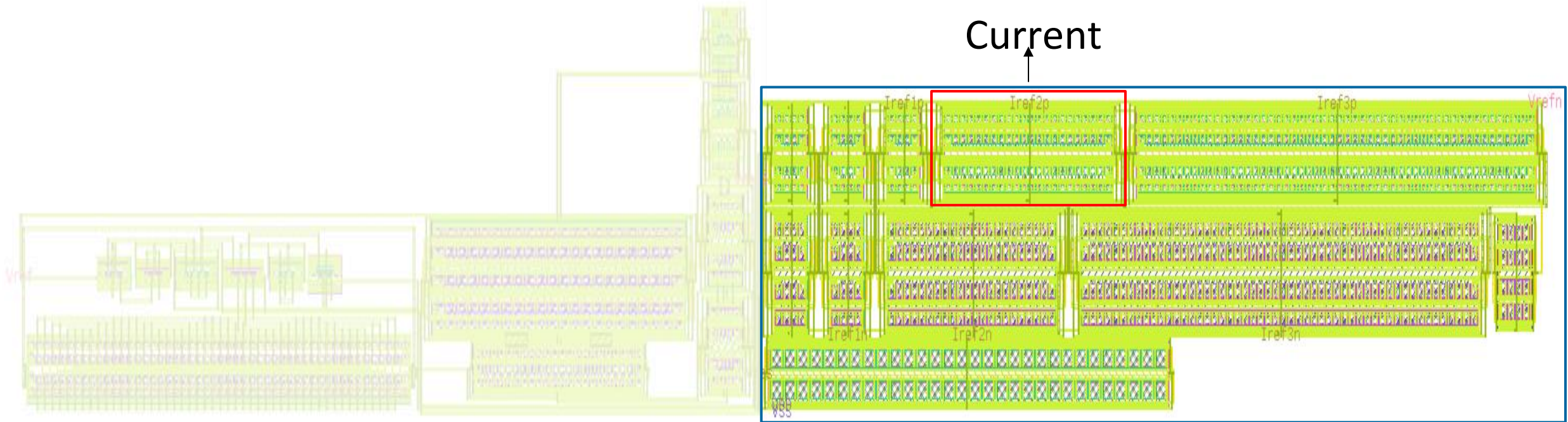
Current Reference layout generated



Current Outputs Circuit

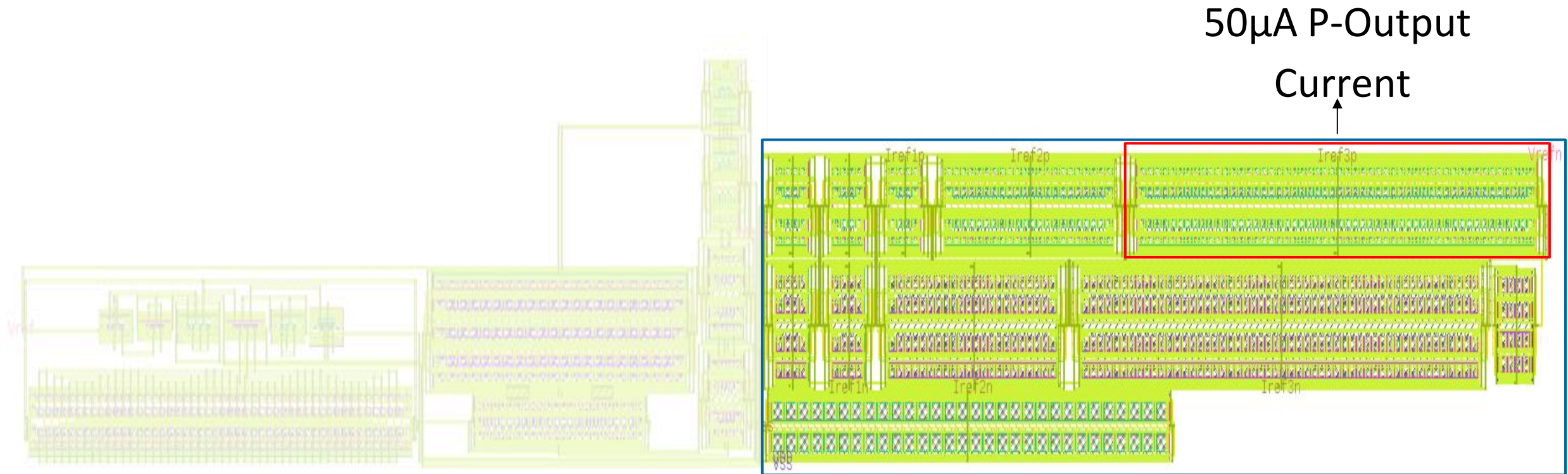
Current Reference layout generated

20 μ A P-Output
Current



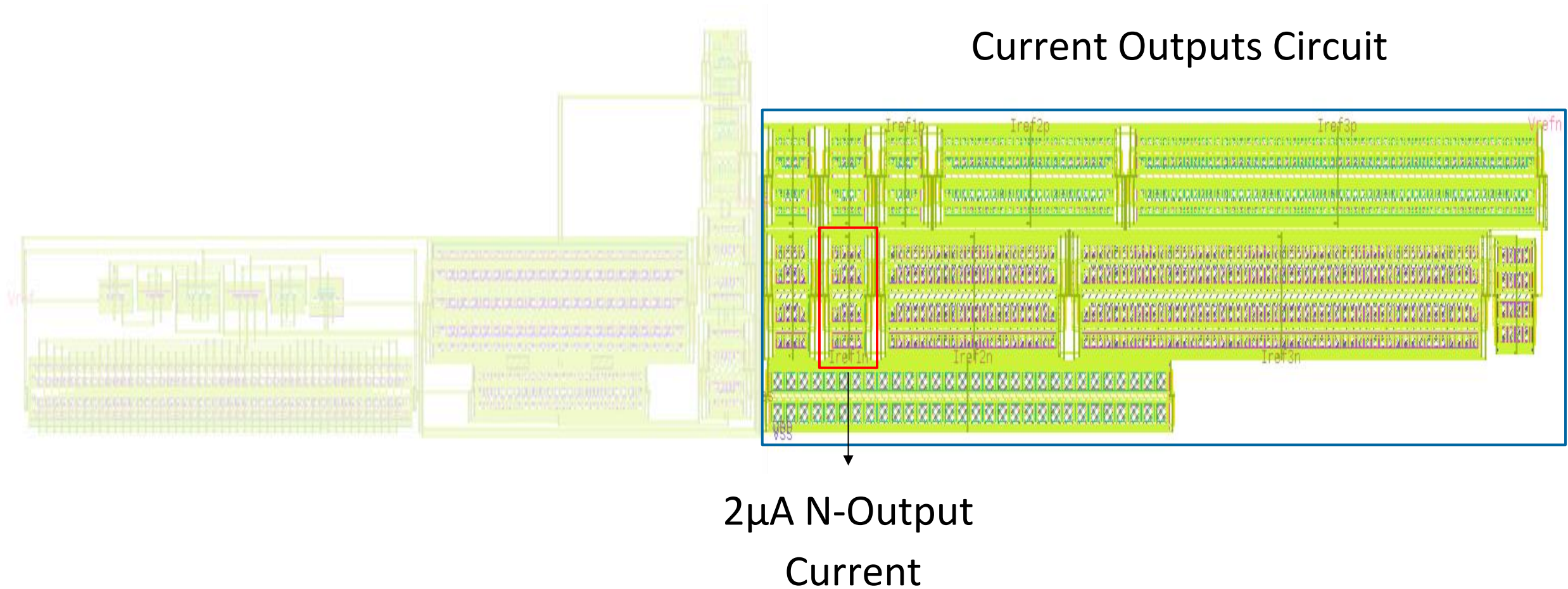
Current Outputs Circuit

Current Reference layout generated

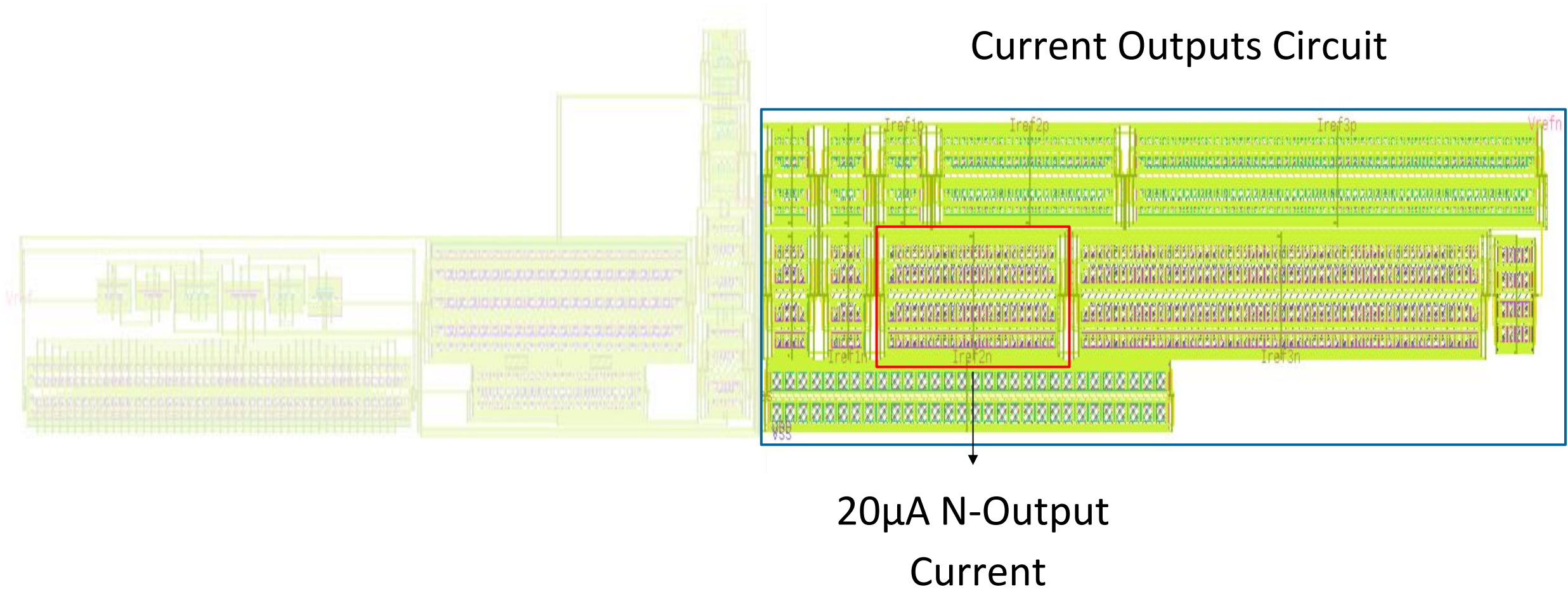


Current Outputs Circuit

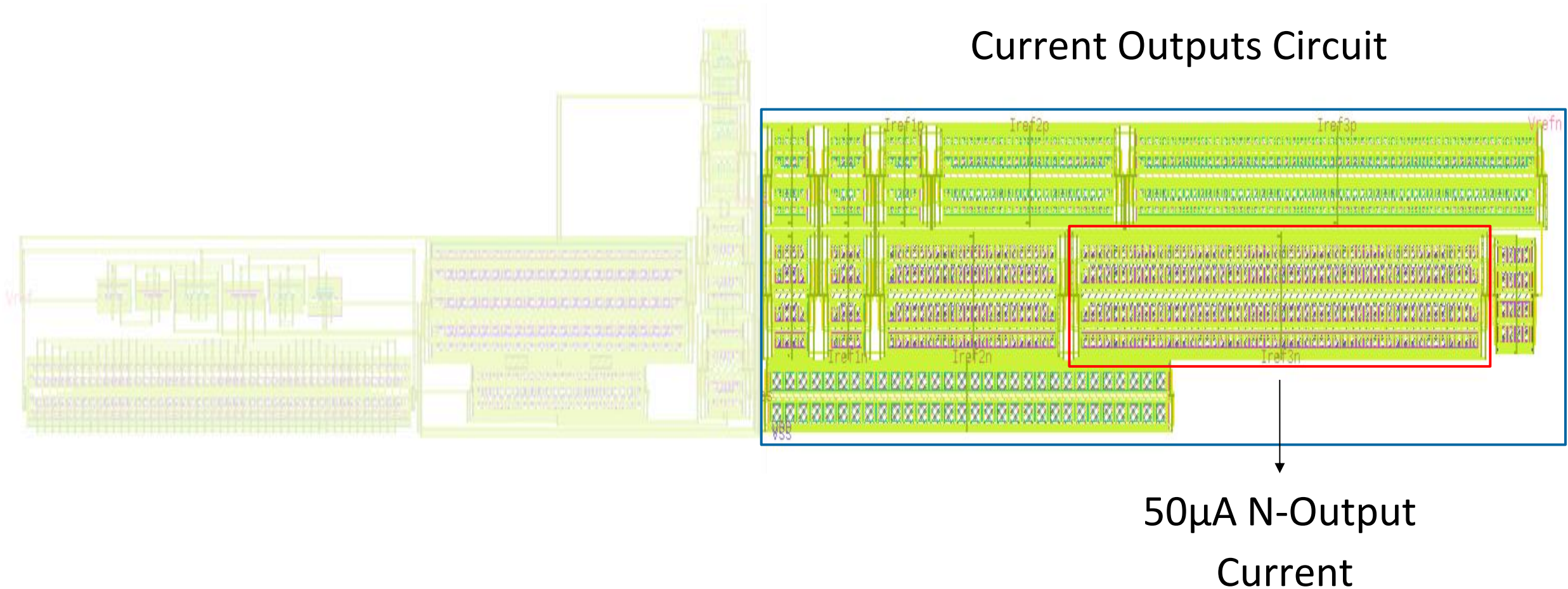
Current Reference layout generated



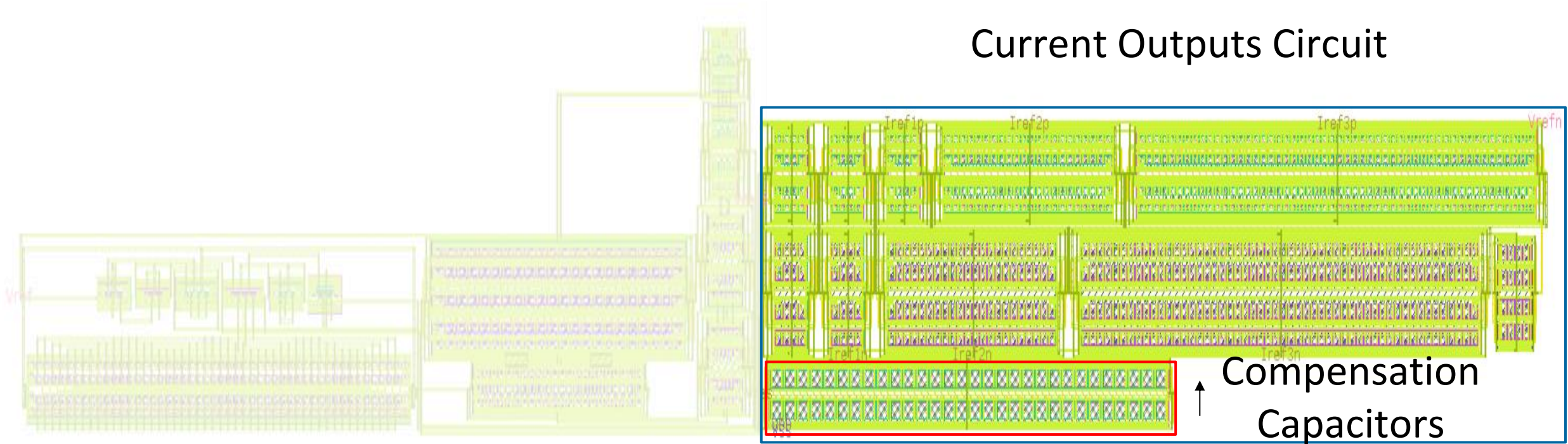
Current Reference layout generated



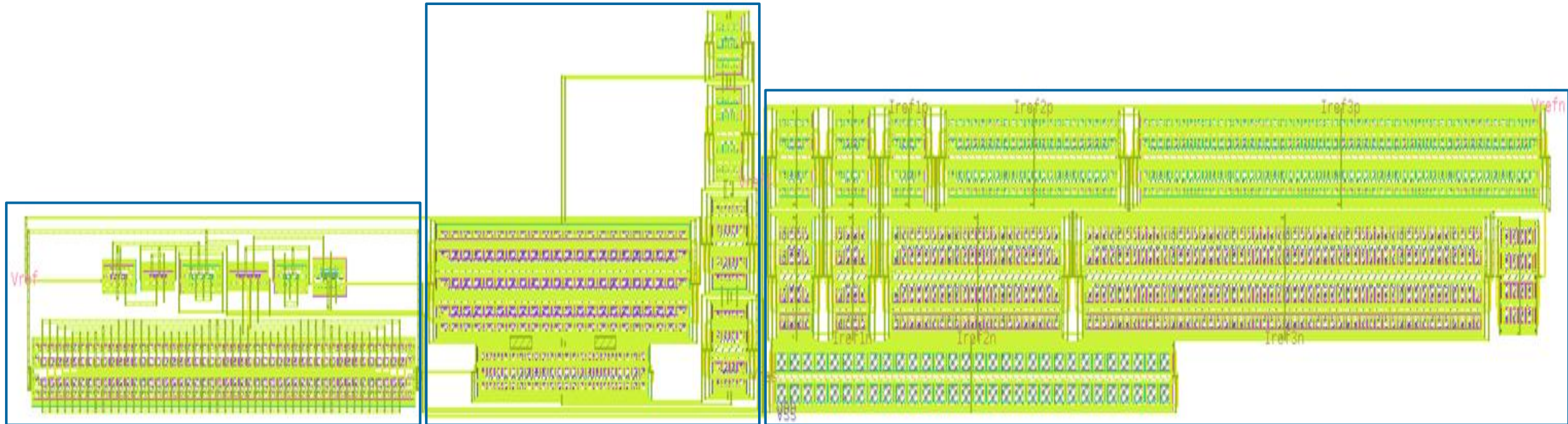
Current Reference layout generated



Current Reference layout generated



Current Reference layout generated



Bias Circuit

Folded Cascode
OTA Circuit

Current Outputs Circuit

Conclusions and Comments

Lessons Learned & Best Practices

Worked fine!

- ▶ We completed the open-source Analog Flow (and more) for **TAPEOUT**.
- ▶ We learned the basics of **Xschem**, **CACE**, **glayout** and **Klayout**, and even **GitHub**.
- ▶ **Template containers** for tool installation and standardization is the best idea!

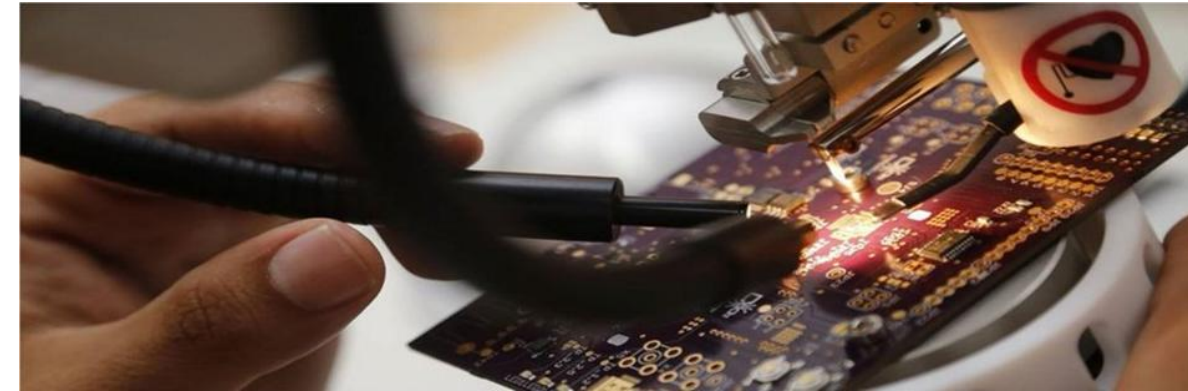
Pending...

- ▶ We didn't use **Magic**.
- ▶ We completed **LVS only for MOS transistors**. Other devices to check.
- ▶ **Documentation** on the expected flow and containers configuration can be improved.

Reflexion and Feedback

- ▶ We noticed the effort made to keep/maintain the **communication** with the working teams (weekly/biweekly meetings, github repo, FossiChat).
- ▶ In the future it would be awesome to explore the possibility of taping out in **sky130**.
- ▶ We understand is a lot of work, but maybe **tutorials** on how to properly develop the expected flow and to use the tools would be awesome.

Thank you SSCS Chipathon Team!



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